Grammaticality de-idealized

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Abstract

This paper introduces the Operator-Value Model of Grammaticality (OVMG), a framework for (un)grammaticality distinct from grammars focused solely on language production or description. The model treats (un)grammaticality as conditioned stability in form–value relations licensed or excluded within communicative situations, separating structural coverage, hard operator-value compatibility, saturation of obligatory dimensions, population licensing, and speaker-level anomaly–confidence read-outs. Previous approaches, including generative grammar, Construction Grammar, and psycholinguistic models, have each captured part of the picture but haven't jointly explained why transparent constructions may nonetheless be deemed ungrammatical, why some lexically odd sentences remain grammatical, why certain transparent structures remain systematically unlicensed, and why categorical grammaticality clusters where it does. The formal core treats grammatical status as the posterior mean of licensed-assembly existence: operator-value compatibility is hard (a unification condition, not a weighted penalty), obligatoriness is derived from preemption of zero-marking rather than stipulated per language, and categoricality emerges as bimodality of the population licensing state under preemption and repair dynamics, rather than from structural membership or intuition alone. The subjective read-out has two channels–anomaly and confidence–which derive the contrast between the confident rejection of preempted gaps and the ineffability of defective paradigm cells. It earns its keep by projection: it predicts which rejected constructions should soften under repeated exposure and which should resist, how learners should treat zero evidence under different opportunity structures, how early second-language rejections should track first-language preemption, how compatibility failures should differ from preempted gaps under change, and where judgment dispersion should rise before means move as a contrast goes moribund.

Keywords: grammaticality; projectibility; acceptability; preemption; entrenchment; conventionalization; usage-based grammar

Introduction

Every competent speaker of English knows that **Can the have running* is impossible, but the source of this certainty proves remarkably elusive. What does it mean to say a sentence is ungrammatical? Consider these examples:

(1) a. **Can the have running?*
b. *Colorless green ideas sleep furiously.* (Chomsky, 1957)
c. **I've finished it yesterday.*
d. ?*I saw Joan, a friend of whose was visiting.*
e. *The bread the baker the apprentice helped made is delicious.*
f. A: *How old are you?* B: **I have 25 years.*
g. **Which did you buy car?*

While all might receive asterisks in many analyses, they represent fundamentally different types of unacceptability. Some constructions, like (1a), fail to establish any form–value relationship in English. Others, like (1c), involve incompatible operator contributions: the present perfect's reference-interval contribution doesn't unify with *yesterday*'s deictic anchoring. Still others, such as (1d), show gradient or indeterminate status. (1e) illustrates what might be called predicted grammatical but reliably rejected: constructions that linguistic theory predicts should be grammatical but that are consistently rejected by native speakers in ordinary judgment tasks. Still others, like (1f), are transparent and grammatical in related languages but lack licensing in English age predication. And a few, like the left-branch extraction of (1g), seem to be ruled out entirely, despite being apparently short and interpretable.

Different traditions emphasize different pieces: formal approaches focus on abstract principles, usage-based theories on frequency and entrenchment, processing accounts on cognitive constraints. Each captures part of the picture, but none unifies categorical constraints, gradient acceptability, cross-linguistic variation, and change over time. I'll argue that the obstacle is a shared idealization, and that removing it reorders the questions.

The de-idealization in the title is specific. Early generative grammar idealized grammaticality to the knowledge of an ideal speaker–listener in a completely homogeneous speech community (Chomsky, 1965, p. 3), treating variation, uncertainty, and change as noise around a categorical competence fact. That idealization bought tractability, but it priced out the phenomena in (1). Following N. Goodman (1955), removing it reorders the questions: ask first what the category GRAMMATICAL lets us predict (its projectibility, what observing some features licenses us to expect about others), and only then what worldly profile supports those predictions and what stabilizes it (Boyd, 1999; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). Mechanisms matter because they explain why the projection holds, not because naming them settles the kind.

This paper develops that answer as the Operator-Value Model of Grammaticality (OVMG). Its unit is the OPERATOR value: a closed-paradigm, update-configuring contrast (tense, agreement, clause type, and their kin), identified by its public-update role rather than by expressive substance, so that tone, particles, word order, and inflection all count when they realize such a contrast (Reynolds, 2026c). The framework rests on three premises, which track the order just named: profile, projection, and support.

1. Profile. Grammaticality is a conditioned form–value relation, specified by whether some assembly covers the form–value pair, whether the assembly's operator contributions unify, whether its obligatory dimensions are saturated, and whether the population licenses the relevant constructions within a communicative situation (community, dialect, register). The profile can fail in distinct ways, while lexical oddness and processing overload can make a licensed form feel anomalous without changing its status, so the phenomena traditionally lumped as "ungrammatical" in (1) fall into separable modes rather than one kind.
2. Projection. The profile earns its standing by what it lets us predict, not by the mechanisms that hold it together: which rejections should soften under repeated exposure and which resist (1d) vs. (1g), how faithfully forms transmit, and how they change over time.
3. Support, graded. The categorical appearance of grammaticality emerges when usage stabilizes the population's licensing state; conversational repair supplies a candidate *controller* for that state, testable by its intervention signature (§4.3); and the subjective read-out of ungrammaticality is a separate, gradient signal–an anomaly–confidence pair–not grammatical status itself.

Together, the premises explain why the traditions have talked past one another. The profile premise preserves the formal insight that constraints can be systematic while letting categorical and gradient cases share one architecture, in line with Francis's (2022) case for building gradience into core grammatical theory. The projection premise keeps the account testable: low-evidence constructions should respond to exposure and framing differently from high-opportunity preempted forms. The support premise explains why formal licensing, processing cost, and subjective read-out can move together without being the same thing.

Section 1 traces the impasse in grammaticality theory to the attempt to treat it as one phenomenon. Section 2 presents OVMG, defining the constitutive conditions and the recurrent instability modes that route a form to one diagnostic category or another. Section 3 gives the state theory: grammatical status is the posterior mean of licensed-assembly existence, with concentration retained for projection. Section 4 gives the dynamics module, showing how such states arise, stabilize, and–where the repair evidence warrants–are actively maintained. Together they derive categoricality and obligatoriness from usage rather than stipulating them. Section 5 situates the account against generative grammar, Construction Grammar, usage-based, logicality, and relevance-theoretic approaches, and states the predictions. The paper closes with limitations and a research program spanning corpus, experimental, and cross-linguistic methods.

1 The impasse in grammaticality theory

The concept of grammaticality remains elusive despite its centrality to linguistic theory. After decades of research, the field still lacks a unified account of what makes an utterance grammatical or ungrammatical. The impasse persists because grammaticality is often treated as a single phenomenon, when judgments actually reflect several questions at once: whether a covering assembly exists, whether its operator contributions are compatible, whether the situation's obligatory dimensions are saturated, whether the relevant constructions are licensed, and how the resulting token feels to a speaker.

1.1 The problem of structural viability

The first challenge in defining grammaticality is distinguishing structures that admit no viable analysis from those that fail for other reasons. Chomsky (1957), building on formal production systems developed by Post (1943), treated grammaticality as a categorical property defined by membership in a set of well-formed strings. That categorical view captures the intuition that some strings, like (1a), resist structural analysis entirely:

(2) **Can the have running?*

Here, the input crashes the structural analysis independent of meaning. The "categorical" view correctly identifies that a viable structural analysis is necessary for grammaticality. But raising this single condition to the *definition* of grammaticality leaves formal approaches struggling with evidence that speakers consistently provide gradient judgments for structures that are clearly analyzable but degraded. The competence-performance distinction (Chomsky, 1965) attempted to preserve categorical grammar by attributing gradience to processing limitations. But as Schütze (2016, p. 71) notes, this often leads to circularity, where problematic results are dismissed as performance artifacts. The OVMG framework resolves this by recognizing that while structural coverage–the existence of some assembly covering the relevant form–value pair–is a necessary, categorical prerequisite, it isn't the sole determinant of grammaticality.

1.2 The problem of operator compatibility

The second tension arises from the interaction of form and value. Chomsky's famous example (1b) shows that operator-valued form–value relations can be intact even when lexical plausibility fails: the sentence is structurally a declarative, the subject is correctly identified both positionally and through agreement, and the tense suggests a general claim. What breaks is the plausibility of the construal, not grammatical status. Conversely, (1c) **I've finished it yesterday* shows that an intended meaning can be transparent while an operator-valued form–value relation fails. Here, the clash is between the present perfect's temporal contribution and the deictic anchoring contributed by *yesterday*.

Generative semanticists like Lakoff (1971) and McCawley (1968) demonstrated early on that many constraints have semantic motivations. Construction Grammar (Goldberg, 1995) has further shown how constructions carry meanings that interact with lexical semantics. Novel uses that align with established constructional meanings (like *She texted him the address*) are accepted, while those that clash with a construction's licensed value (like **She disappeared him the evidence*) are rejected. A complete theory needs a hard compatibility condition for conflicts among grammatically encoded operator contributions, while keeping lexical implausibility and processing cost in the subjective cost vector rather than turning every odd construal into a status failure.

1.3 The problem of situational licensing

Finally, even structurally viable and semantically transparent forms may be ungrammatical if they violate community conventions. Sociolinguistic research (Labov, 1972) demonstrates that grammaticality references community norms. Cross-linguistic variation shows that each language community conventionalizes particular form–value mappings through historical processes. These conventions become entrenched through statistical preemption (Goldberg, 2011): speakers learn that certain forms are ungrammatical precisely because they encounter alternative expressions in contexts where the ungrammatical form would otherwise be expected.

Consider (1f), repeated here:

(3) A: *How old are you?* B: **I have 25 years.*

This utterance is structurally viable and semantically transparent (the intended meaning is clear), but it's categorically rejected in English. The same form is grammatical in French (*J'ai 25 ans*) and Spanish (*Tengo 25 años*); the constraint is community-specific. Usage-based approaches (Bybee, 2006) emphasize that such rejection arises from the absence of situational licensing. The English-speaking community has conventionalized a different way to express age, so the *have*+age frame is preempted by the BE-frame in a dense niche rather than merely being unattested.

These problems don't reduce to one scale. Structural coverage, operator compatibility, saturation, licensing, and subjective read-out have different diagnostics and different evidence streams. Acknowledging them as distinct but interacting routes makes it possible to move beyond the impasse of trying to force all grammaticality judgments into a single "competence" definition or a purely probabilistic usage model.

2 The Operator-Value Model of Grammaticality

The term VALUE is used here in the Saussurean sense of *valeur* (Saussure, 1916): the identity of a linguistic unit is constituted not by intrinsic properties but by its position in a system of contrasts, what it patterns with, what it opposes, and what interpretations it makes available. This isn't feature-value assignment or decision-theoretic value. Throughout, I treat grammatical knowledge as a conditioned form–value relation; when this relation is sufficiently stable in a communicative situation, with a dominant value reliably recoverable and socially licensed, I refer to it informally as an established form–value relationship.

The model's name marks its unit: operator values (closed-paradigm, update-configuring contrasts). A companion operator-stratum paper develops this boundary in full and argues that it follows public-update role rather than expressive substance: tone, particles, word order, inflection, and suprasegmental material all fall within its scope when they realize an operator contrast (Reynolds, 2026c).

Here a self-contained working test suffices, because the concept has to stand on its own inside this paper: it fixes the scope of hard compatibility, the categoricality prediction, and the appendix on Turkish suffix harmony (§A).

A contrast is an operator contrast when it meets three conditions jointly: drawn from a *closed* paradigm, *high-opportunity* (the communicative job recurs constantly), and *configured to the public update* (clause type, polarity, tense and aspect anchoring, agreement and role allocation, scope, or reference tracking). Mis-setting such a contrast changes the update instruction even when the intended message is recoverable. The three conditions have formal counterparts in §§3–4: closure is finiteness of the dimension's value set, opportunity is the niche count $N_t(n, c)$, and update-configuration is a positive expected common-ground divergence $\Delta(d) > 0$ from mis-setting the contrast (Reynolds, 2026c). $\Delta(d)$ needn't be identified from repair itself: comprehension probes can ask whether mis-setting d changes hearers' answers about who did what to whom, when, with what force, or under which evidential commitment.

These conditions exclude cases that "closed-class", "grammaticalized", or "high-frequency" alone would admit. A closed-class item that doesn't configure the update, such as a purely stance-marking discourse particle, fails the third condition and is predicted *not* to attract categorical policing; a high-frequency open-class collocation fails the first; and a genuinely update-configuring contrast that's

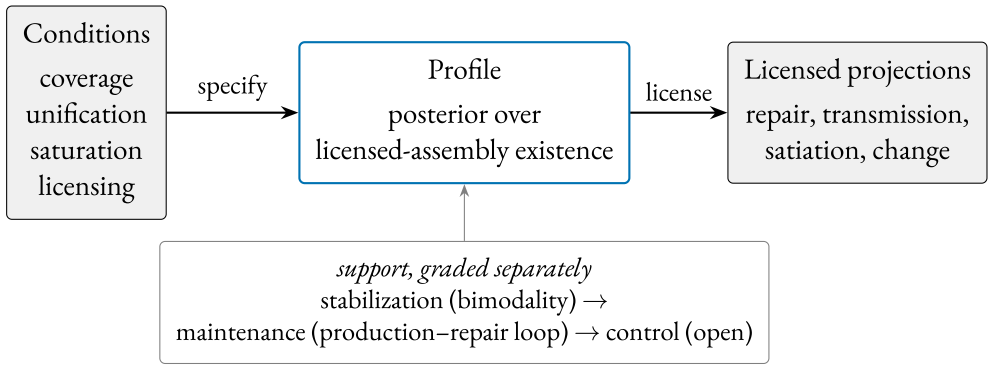

Figure 1: The projectibility-first order of explanation. The constitutive conditions specify a licensed-assembly profile whose posterior mean and concentration license the projections on the right. What stabilizes, maintains, or controls the profile is a separate, graded claim (below), not something inherited from naming the mechanism.

gone low-opportunity, such as a moribund case distinction, fails the second and is predicted to destabilize. The prediction is falsifiable in the direction that matters: a closed, high-opportunity contrast that isn't update-configuring should be policed like style, not like grammar.

The formal architecture rests on four constitutive conditions: structural viability, operator-value compatibility, saturation of obligatory dimensions, and situational licensing. Together they specify a PROFILE in the projectibility-first sense: a pattern whose partial observation licenses further expectations (Reynolds, 2026a). Their interaction determines grammaticality (§3); what the profile projects, and what supports those projections, is the business of §3 and §4. Figure 1 lays out this order of explanation, which the rest of the paper fills in. Table 1 summarizes these conditions; the formal definitions appear in §3.

Condition	Component
$\mathcal{A}(f, v) \neq \emptyset$	Structural coverage: some value-matched assembly exists
$\mathrm{def}(A)$	Operator-value compatibility (unification)
$\mathrm{sat}(A, c)$	Saturation of obligatory dimensions (derived, §4.5)
$L_t(A, c)$	Predictive assembly licensing from node-level θ_t

Table 1: The conditions used by the OVMG framework. Saturation is derived from the licensing state rather than an independent primitive. Their conjunction, existentially quantified over covering assemblies and read through the learner's posterior, gives grammatical status (§3.3.1).

The formal core needs names for recurrent INSTABILITY MODES: ways of defeating coverage, unification, saturation, or licensing, with processing and prescriptive factors modulating how grammatical something *feels* without directly fixing grammatical status.

2.1 Diagnostic categories

The categories below are labels for common routes to ungrammaticality, not primitives of the model. Each traces to one or more constitutive conditions, as the formal core in §3 makes explicit.

Form–value relations. Grammaticality depends on stable links between forms and the operator values they realize within communicative situations. That form–value relations exist at every level of grammatical organization, including morphemes and abstract syntactic patterns, is a core insight of Construction Grammar (Goldberg, 1995). OVMG narrows the target of categorical grammaticality to closed-paradigm, high-opportunity, update-critical contrasts: resources that configure how an utterance enters public coordination.

Operator exponents, like words, are typically polysemous: a single form participates in multiple form–value relationships, usually with one dominant value (Kilgarriff, 2004). The English past tense, for instance, usually denotes past time, but it can also mark social distance (*Could you help me?*) or low likelihood (*If I went tomorrow...*). What matters for grammaticality is whether a stable relationship exists within the relevant communicative situation, not whether the form has a single fixed meaning.

Neuroimaging work provides converging evidence that form and meaning are not processed by wholly sealed modules. Fedorenko and colleagues distinguish language-selective regions from neighbouring domain-general systems while showing that syntactic and semantic manipulations both recruit the language-processing architecture (Fedorenko et al., 2011, 2012, 2024). The processing claim here is modest: the subjective read-out can reflect interacting processing demands even when grammatical status remains intact.

Following Wiese (2023, p. 3), grammaticality is conditioned by regularities within a "communicative situation". The emergence claim defended below is narrower: categoricality arises through the licensing dynamics in §4.4, not through an unanalyzed appeal to usage.

Communicative situations aren't static or mutually exclusive. A speaker orients to overlapping contexts that shift on multiple dimensions: the immediate situation (formal meeting vs. casual chat), durable social positions (regional, generational), and the norms they're orienting to (which can shift with audience). This fluidity explains style-shifting, dialect formation, and the gradual acceptance of initially marginal constructions.

Early child language shows this. Toddlers produce utterances that deviate from adult norms but remain internally consistent within the child's developing system. A toddler in a monolingual English household might say:

(4) *Ava cookie.* (intended as 'Ava = I want a cookie')

Although this differs from adult English norms, it may not be perceived as ungrammatical by the child's regular caregivers. Used with the same pragmasemantic force among anglophone adults, the construction would be judged ungrammatical.

Multiple modal constructions provide another clear example of variety-relative grammaticality:

(5) *I might could help you with that.*

This combination of modal auxiliaries is systematically possible for some American English speakers, who can productively generate similar constructions (Morin & Grieve, 2024). Speakers from communities where only single modals are grammatical, though, typically reject such combinations as ungrammatical.

A similar pattern appears in code-mixing among bilingual speakers, where combinations of forms from different languages can be grammatical within that bilingual community's norms but not without. Consider a Spanish-English bilingual speaker who uses a Spanish progressive auxiliary with an English lexical verb:

(6) *Ayer, estábamos lifting en el gym durante una hora.*
yesterday be.IMPF-1PL lifting in the gym for an hour
'Yesterday, we were lifting in the gym for an hour.'

Within the right communicative situation, this utterance is grammatical. The Spanish auxiliary *estábamos* combines with an English participial form *lifting* to form the progressive aspect. This cross-linguistic combination of morphology and a lexical verb is consistent with local norms, where code-mixed utterances are common and meaningful. In contrast, in a standard monolingual Spanish setting, where fully Spanish progressive structures are licensed (*estábamos levantando pesas*), speakers may judge the example in (6) as ungrammatical. The use of intransitive *lifting*, specific to the gym community, further illustrates just how localized grammaticality judgments can be.

This doesn't mean bilingual communities simply accept any combination of languages. As Toribio (2001) reports, Spanish–English bilingual speakers judge examples like (7) as unacceptable, showing that even in bilingual communities, there are systematic constraints on which language combinations are permitted.

(7) * *Los enanitos intentaron pero no succeeded in awakening Snow White* (Toribio, 2001)
the dwarfs try.PST-3PL but not succeeded in awakening Snow White
'The dwarfs tried but didn't succeed in awakening Snow White.'

In a slightly different case, as a second-language speaker of Japanese, I used to say

(8) * *Kawaii da.*
cute COP.NPST
'(That)'s cute.' (intended)

Conventionally, though, plain *i*-adjective predicates in Japanese do not take the plain copula *da* in this frame. The issue is exponent selection: the predicate's value is recoverable, but the community does not license that copular exponent there. The example shows why acquisition history matters: shared routines stabilize form–value relations for a community, while individual trajectories can still pull judgments away from that norm.

The specific linguistic context can matter too:

> To take an obvious case which Jerry Morgan [(1973)] discussed recently, certain combinations of words are extremely strange if presented in isolation but are perfectly normal as answers to certain questions. *Spiro conjectures Ex-Lax* would generally be felt to be unintelligible if presented out of context but is a perfectly normal answer to the question *Does anyone know what Mrs. Nixon frosts her cakes with?* (McCawley, 1974, 252, italics added)[1]

These examples show why grammaticality has to be indexed to communicative situation. Conversants who differ in immediate situation, social position, or the norms they orient to may disagree on whether a form is grammatical, and both may be right within their respective contexts. The linguist's task is to determine whether a form is grammatical from some position in that space, or systematically excluded from all of it.

[1] The humour derives from political tensions of the Watergate era, when both Vice-President Spiro Agnew and President Nixon would ultimately resign from office.

The stability of form–value relations within speech communities can be empirically modeled, as demonstrated by Blythe and Croft (2009). Their utterance selection model treats speech communities as networks where speakers track and reproduce linguistic variants based on their interactions. When applied to dialect formation, these models show how competing linguistic variants spread and eventually stabilize, with initial variant frequency strongly predicting which form will prevail.

In different languages, different distinctions become conventionalized as grammatically obligatory. These obligatory contrasts reflect historically stabilized patterns in which meaning distinctions are systematically marked. Over time, usage establishes grammatical constructions that reliably signal these distinctions. As a result, what counts as a grammatical necessity in one language may be optional or absent in another.

The progressive aspect provides a useful example. In Standard English, it isn't optional but an obligatory grammatical marker for ongoing, incomplete actions:

(9) *She is studying right now.*

Here, the progressive form *is studying* isn't just a stylistic choice. It's the recognized, grammatical way to convey the appropriate aspectual meaning in this frame.

In French, by contrast, the progressive aspect isn't a grammatically mandatory distinction. Speakers can signal ongoing activity through adverbs or periphrastic constructions, but standard French doesn't have a dedicated progressive form. The sentence:

(10) *Elle étudie maintenant.*
'she studies/is studying now'

can comfortably describe a currently ongoing action without any need for special morphology. In Standard French, this aspectual distinction isn't conventionalized as an obligatory operator contrast. English grammar treats the contrast as obligatory; French doesn't. On this account that difference isn't a primitive of either grammar: §4.5 derives obligatoriness as the limit of preemption applied to zero-marking, so the English–French contrast is a prediction of the dynamics, not a stipulation they inherit.

The same holds for evidentiality, the grammatical marking of information sources. In Turkish, evidentiality is systematically realized through verb forms and particles that distinguish between directly witnessed events and those inferred or reported:

(11) *Gelmiş*
'He/she came (apparently)'
(i.e. the speaker wasn't a witness but inferred or heard about it.)

In Turkish, evidential distinctions are conventionalized as obligatory operator contrasts. A speaker can't simply omit evidential marking without sounding ungrammatical.

English speakers can say *I heard that he arrived* or *He must have arrived*, but these are optional lexical or modal resources rather than required elements of the grammar. In English, evidential distinctions aren't conventionalized as obligatory operator contrasts. The same derivation covers the Turkish case on a different dimension (§4.5).

Kilani-Schoch and Dressler (2005) demonstrate this principle systematically in their analysis of verbal morphology across French and other Romance languages. They show how apparently similar verbal systems can grammaticalize quite different semantic distinctions as obligatory. The contrast reflects different conventions about which meaning distinctions are systematically marked. For instance, while both French and Italian mark aspect morphologically, they differ in which aspectual distinctions are grammaticalized versus left to optional lexical expression.

Establishing a form–value mapping is necessary for grammaticality, but it isn't sufficient. Speakers routinely interpret strings that still count as ill-formed in the relevant communicative situation. That flexibility makes licensing, not interpretability in principle, the central question.

The conditions do different work. Coverage asks whether any assembly can be built; compatibility asks whether its operator contributions unify; saturation asks whether the obligatory dimensions for the situation are set; licensing asks whether the population treats the relevant constructions as repertoire. Processing costs, recovery costs, and lexical implausibility can still make a licensed assembly feel bad, but they enter selection and the subjective cost vector rather than replacing status. The interaction predicts why **Furiously sleep ideas green colorless* elicits universal rejection while (1e) feels bad in everyday quotation but is accepted as "grammatical but hard to process" once speakers are walked through the intended structure.

Nor does convergent usage by itself settle the matter. Defining grammaticality as "whatever speakers accept" would be circular: it couldn't flag contested innovations, explain stable dialectal differences despite mutual intelligibility, or distinguish licensed forms from performance errors that pass unnoticed. Separating coverage, compatibility, saturation, licensing, and subjective read-out avoids trivializing grammaticality while still treating durable population-level licensing as decisive for which relationships endure.

Operator-value compatibility. Beyond the existence of form–value relations, grammaticality requires compatibility among the operator values those relations realize. This condition captures cases where individual elements are well formed but their combination creates conflicts among grammaticalized contrasts activated in the communicative situation.

Consider the temporal incompatibility in:

(12) * *I've finished it yesterday.*

Here, the present perfect construction contributes a reference interval anchored to present relevance, while *yesterday* contributes a deictically past interval. Unlike lexically incongruous combinations like *colorless green ideas*, which remain grammatical because the clash doesn't mis-set an operator value, this example shows failed unification between grammatically encoded temporal contributions.

The age example belongs elsewhere:

(13) * *I have 25 years.* intended as 'I'm 25 years old'

In English, *have+years* is a licensed frame for duration, remaining time, or experience, but ordinary age predication is licensed through a different frame. The intended value is transparent, so the failure isn't an operator-compatibility failure. It's a licensing failure for an age-predication construction in English, even though cognate-looking frames are licensed in French and Spanish.

Sometimes compatibility failures involve conflicting information-packaging requirements:

(14) * *Who did the lifeguard who saved _ work in New Jersey?* (Cuneo & Goldberg, 2023)

The clash is in grammatically encoded information packaging: the same participant is simultaneously focused through fronting *who* while being backgrounded by the relative clause construction (Cuneo & Goldberg, 2023).

These examples also delimit compatibility. The formal core treats operator-value conflict as failed unification among partial operator assignments: tense–aspect anchoring where tense–aspect is obligatory, argument-linking requirements where a construction licenses roles, information-packaging instructions where a construction grammatically encodes focus or backgrounding, and indexical contrasts only when they've entered a closed accountable repertoire. This makes compatibility partly derivative of licensing facts at the feature level, keeping arbitrary lexical implausibility out of grammatical incompatibility.

Processing constraints. Language processing engages both dedicated language networks and domain-general cognitive systems (Fedorenko et al., 2024). When constructions overload these systems, particularly through multiple long-distance dependencies or heavy embedding, they may trigger feelings of ungrammaticality despite being structurally well-formed.

(15) *The bread the baker the apprentice helped made is delicious.*

Evidence for these constraints comes from multiple sources. Studies of sentence processing show that dependencies spanning multiple intervening elements increase processing difficulty, with new referents between dependent elements compounding memory load (Gibson, 2000, 2026). In (15), while each individual relation (like *the apprentice helped* and *the baker made*) is interpretable in isolation, their nested combination overwhelms incremental processing. The construction is grammatical, but excessive bridging costs from its nested structure trigger feelings of ungrammaticality.

Rather than reflecting simple memory limitations, processing constraints emerge from the interaction between specialized language networks and other cognitive systems (Fedorenko et al., 2024). On this processing-level interpretation, what speakers experience as difficulty may reflect demands distributed across specialized brain systems rather than a single cognitive bottleneck.

Dependency locality is the best-documented such cost. Integration difficulty rises with dependency length and peaks at loci of long-distance integration, with locality accounting for substantial region-level variance in reading times in Grodner and Gibson (2005). Because the effect is robust across processing work, the formal treatment in §3 gives dependency locality a dedicated component rather than hiding it inside a residual term. High-locality constructions feel worse even when structurally well-formed, and because they're harder to produce and parse, speakers tend to avoid them.

That avoidance is a pathway from processing to change, developed formally in §2.5 and §4: lower-processing forms are easier to store and reuse, so simpler patterns can spread and conventionalize. The processing evidence supports this as a plausibility claim about one route into stability, not a direct neural explanation of grammatical change.

Processing constraints also interact with the licensing state without becoming part of status. A construction that is marginal because its licensing posterior is diffuse may be judged more harshly when it is also hard to process. Conversely, a highly licensed construction can remain grammatical despite considerable processing demands, even while those demands keep its anomaly signal high.

Socio-pragmatic indexicality. The value of a construction extends beyond semantic content to include socio-pragmatic dimensions: how linguistic forms index aspects of social context, speaker identity, group membership, stance, or interpersonal relationships (Eckert, 2012; Silverstein, 1976).

In many varieties of Latin American Spanish, for example (e.g. the Río de la Plata region), speakers use *vos* and its associated verb forms instead of the *tú* forms used elsewhere in the Spanish-speaking world (Bertolotti, 2016):

(16) ¿*Vos* *querés* *un café?*
you.SG want.2SG-VOS a coffee
'Do you want a coffee?'

Here, the use of *vos* rather than *tú* denotes the second-person singular hearer (the person being offered a coffee) and indexes the speaker's regional identity and familiarity with the local dialect. This indexical value may convey closeness, solidarity, or membership in a particular geographic and social community.

This situational view of grammaticality can manifest asymmetrically. Speakers from the Río de la Plata region may view a conversational situation as accommodating both their own norms and those of *tú*-using interlocutors; the situation itself can encompass both *vos* and *tú* as grammatical options. But speakers from *tú*-only regions might conceptualize the same situation more restrictively, defining it in a way that categorically excludes *vos* as a grammatical possibility. This asymmetry doesn't depend on different understandings of the forms themselves, but can arise from different ways of defining what the communicative situation allows, influenced by the indexical values attached to *vos* versus *tú* in their respective communities.

Phonology can carry constructional indexical value, but it bears on grammaticality only when it realizes an operator contrast or enters an operator-exponent licensing condition. Babel (2025) provides an example. In a study conducted in Bolivia, participants were presented with audio stimuli where only the vowels were manipulated to reflect either a highland or lowland accent. This presented participants with incongruent identity cues (e.g. vowels from one accent along with consonants from the other). But this didn't trigger feelings of ungrammaticality. Instead, vowel contrasts activated expectations about consonant features and discourse markers, and some participants "hallucinated" identity-linked features that weren't present in the signal.

So a construction's value reaches beyond semantic features to socio-pragmatic aspects that shape how speakers and hearers negotiate authority, identity, solidarity, and other interpersonal relations. Recognizing these indexical dimensions helps explain why certain forms feel natural and grammatical to some speakers and out of place or even ungrammatical to others.

SITUATIONAL LICENSING. Even well-formed constructions with clear meanings may be judged ungrammatical if they lack conventionalized licensing. This condition concerns whether a form–value relation is licensed within the relevant communicative situations.

(17) * *We sheared three sheeps.*

The regular plural marking is semantically transparent and structurally parallel to other English plurals, but the irregular form *sheep* is entrenched across English-speaking communicative situations, making the regularized version unacceptable. Without a metaphorical or playful justification (as in *the black sheeps of the family*, where the irregular plural might signal a figurative usage), the utterance is ungrammatical.

Situational licensing operates independently from the other conditions. A construction may have complete operator compatibility and minimal processing demands but still be excluded because a different form has been conventionalized for that meaning. Extreme rarity makes the distinction clear. Some constructions are so infrequent that speakers lack a shared consensus about their status.

The independent relative genitive pronoun *whose* provides an example, being so unusual that Hankamer and Postal (1973) deem it non-existent. The contexts that license this construction require the simultaneous convergence of distinct pragmatic and syntactic conditions (sufficient accessibility of both possessor and possessum, the appropriate information structure, and an environment allowing ellipsis), which are seldom met all at once (Reynolds, 2024):

(18) ? *I saw Joan, a friend of whose was visiting.*
(adapted from Huddleston & Pullum 2002, p. 472)

In the COCA check reported here, no token of the construction appears (Davies, 2024). That absence supplies little negative evidence, because the opportunity set is tiny: contexts requiring an independent relative genitive are vanishingly rare. Speakers have little evidence either way: neither positive tokens nor the systematic absence that would accumulate if the construction were regularly passed over. The evidence supports high *uncertainty* rather than confident rejection. Some speakers can draw an analogy with similar constructions and accept examples like (18); others remain unsure; still others, lacking sufficient exposure, can't construct a stable form–value relation at all. Even among those who grasp the construction analytically, Shain et al. (2020)'s account predicts that its unexpectedness would trigger high surprisal, compounding the uncertainty with processing difficulty.

This contrasts with putative preempted cases like left-branch extraction, discussed next, where the opportunity set appears large and the construction is systematically untaken. The classification depends on a further quantity, though: how often speakers would choose the discontinuous form if it were stipulated to be licensed.

Apparent categorical exclusions. Some constructions face rejection so persistent that it *appears* categorical. Left-branch extraction is the textbook case:

(19) * *Which did you buy* [__ *car*]*?*

The intended meaning is easily grasped; nothing in the semantics or pragmatics prevents understanding the speaker's intent. Nor does the construction seem excessively complex to process. Yet English speakers categorically reject it, and Snyder (2022, p. 22) treats this kind of pattern as a core case of "ungrammaticality".

Under OVMG, such patterns are candidate preempted gaps: situational licensing is driven toward zero by persistent preemption when the communicative job is common, an established competitor wins, and the counterfactual choice probability of the missing form is high. For English left-branch extraction, the natural competitor is *Which car did you buy?*. The corpus absence is suggestive but not decisive; the missing ingredient is independent norming of $\rho^\star_t(\textsc{lbe} \mid n, c)$ under permissive framing. Additional processing penalties, such as systematic garden-path reanalysis when the bare interrogative phrase is encountered, can strengthen the subjective categoricality without any independent structural constraint. The formal dynamics appear in §4.

Cross-linguistic comparison doesn't threaten this analysis, because the dynamics are defined over a language's own niches. Left branch is a descriptive class tied to a particular phrase-structure analysis, not a comparative concept (Haspelmath, 2010); whether discontinuous nominals in case-rich, article-less languages instantiate the same configuration as the English case is far from clear, and the companion corpus study deliberately brackets the comparison (Reynolds, 2026d). The model instead predicts which opportunity structures should support similar patterns. Discontinuous nominal orders should be learnable where inflectional marking makes the dependency recoverable, a productive discontinuous-order pattern supplies scaffolding, and a discourse niche makes the separated element useful. English meets none of the three, so the candidate has no scaffolding of its own and the contiguity-preserving competitors win every opportunity.

Several diagnostic criteria help identify these preempted-gap constructions:

1. Persistent non-licensing: The construction doesn't improve under ordinary exposure or framing while the high-concentration non-licensing state remains in place.
2. Categorical rejection: Speakers find the constructions simply impossible, with high confidence rather than mere uncertainty.
3. Resistance to satiation: Unlike processing-heavy cases, repeated exposure doesn't improve acceptability.

Prediction: If LBE is really a high-concentration preempted gap, framing tokens as intentional dialectal or ingroup resources should do little to move judgments; the same manipulation should matter more for low-concentration cases such as independent relative *whose* or defective cells. A robust framing effect for LBE, or a low independently normed $\rho_t^\star$, would count against the preempted-gap diagnosis.

If no covering assembly is available at all, the diagnosis isn't a further residual constraint but structural coverage failure: the existential over assemblies is empty. Treating classic cases like LBE as driven to zero under preemption keeps the constitutive core minimal and pushes the explanatory work into the dynamics of situational licensing, while reserving true coverage failure for cases with no well-typed route through the constructional inventory.

WINNERLESS CELLS: PARADIGM DEFECTIVENESS. Preempted gaps have a winning competitor. Defective paradigm cells don't. They leave a high-opportunity cell empty without any single form winning it. A clearer case is the Russian first-person singular non-past of *pobedit'* 'win': speakers avoid the expected cell and resort to periphrasis instead. The English *amn't* gap is diagnostically useful but less clear, because broader niches include *I'm not* as a competitor, while narrower contracted-negator niches make the cell look winnerless (Broadbent, 2009; Pertsova, 2016; Pertsova & Kuznetsova, 2015; Sims, 2023; Thoms et al., 2025).

In the Russian case, the niche is common and candidate forms are transparent, but no candidate accumulates enough positive evidence to settle as the form. The source can be structural uncertainty, lexical restriction, learning dynamics, or avoidance, with the mixture varying across cases (Sims, 2023). In this account, defectiveness is a third licensing profile: high opportunity with evidence divided among candidate forms, leaving each candidate's posterior low in mean but unstable, rather than concentrated near zero.

The phenomenology matches: speakers report not knowing how to say it (ineffability), not the confident "that's wrong" of a preempted gap. §4.6 derives this profile from the production model itself: avoidance absorbs the cell's opportunities, and because avoidance is nearly uninformative about any particular candidate, no candidate accumulates confident negative evidence either. The satiation-framing prediction follows: defective cells should behave like low-evidence forms, movable under framing, not like preempted gaps.

2.2 Diagnosing (un)grammaticality at a glance

For exposition the recurrent instability modes can be read as a short decision tree (the formal definitions appear in §3):

Condition	⟶ Canonical outcome
no covering assembly	⟶ NONSENSE (**Can the have running*)
operator-value clash (unification failure)	⟶ compatibility failure (**I've finished it yesterday*)
obligatory dimension unset	⟶ saturation failure (unmarked progressive in English)
rarely encountered, low certainty	⟶ community-novel (*friend of whose*)
systematically preempted, high confidence	⟶ preempted gap (left-branch extraction if normed)
high opportunity, candidates split	⟶ defective cell (*pobedit'* 1SG)
high processing cost, structure recoverable	⟶ transiently ill-felt (centre embedding)
otherwise	⟶ grammatical

Lexical-semantic oddity without an operator clash (*colorless green ideas*) routes to "grammatical"; its strangeness is a construal-plausibility cost in the subjective read-out (§3.7), not a status fact.

2.3 Patterns of (un)grammaticality

These categories aren't points on one severity scale. They name different routes through the same architecture. A token can fail because no assembly covers it, because its operator contributions don't unify, because an obligatory dimension is unsaturated, or because the pivotal construction isn't licensed in the population. Separately, a licensed token can feel bad because processing, plausibility, or prescriptive dissonance pushes the anomaly channel away from zero; confidence tracks how settled the relevant status condition is.

This matters for apparent gradience. Intermediate acceptability ratings can come from a diffuse licensing posterior, from a strong anomaly signal attached to an otherwise licensed form, or from disagreement about the conditioning state. They aren't automatically evidence for soft compatibility. Semantic transparency can make a speaker recover the intended value, and it can affect attribution or repair, but it can't make an operator clash grammatical. Conversely, high processing cost can make a licensed construction feel bad without lowering its licensing posterior. The formal core in §3 separates these routes; the worked examples in §3.9 show how they project differently.

2.4 The subjective experience of grammaticality

The constitutive conditions determine conventional grammatical status (§3.3); speakers register putative violations through read-outs and judgments that can diverge from it. [2]

THE SUBJECTIVE READ-OUT. The familiar "feeling of ungrammaticality" is the anomaly side of a broader metacognitive read-out. It parallels other well-studied metacognitive feelings like the FEELING OF KNOWING (FOK; Hart, 1965), which arises when one feels certain of knowing something but can't retrieve it.

[2] The distinction between grammatical status and the subjective read-out has a philosophical analogue in Cavell's (1969) distinction between knowledge and acknowledgment. For Cavell, acknowledgment isn't a separate capacity from knowing but an inflection of it, bearing knowledge toward the situation that produced it. Speakers don't merely *know* that a string is ungrammatical; they *register* it as a violation, with metacognitive consequences. For LLM-based grammaticality research, the distinction blocks an easy inference: LLMs can produce judgments that correlate with human ones, but whether they acknowledge the norms they're applying or merely process patterns that correlate with those norms remains an open question (Techio, 2026).

There isn't a positive feeling of grammaticality, just as there isn't a feeling of having enough oxygen, only the negative feeling registered in its absence. The anomaly channel, then, is the negative response triggered when an utterance violates expected form–value patterns. The confidence channel tracks how settled the speaker takes the relevant status condition to be.

This notion echoes Edward Sapir's "form-feeling", an often unconscious grasp of language patterns (Sapir, 1921, 1927). Evidence from aphasia supports the same separation. Patients with Broca's aphasia, who produce agrammatic speech, often exhibit self-monitoring behavior, attempting self-correction and expressing frustration with their grammatical errors (Oomen et al., 2005).

Unstable or missing form–value relations, detected during processing, can trigger this response. The response is a speaker-level heuristic for detection, not the definition of grammaticality. That separation explains why marginal constructions elicit gradient certainty, why repeated exposure can attenuate negative responses without necessarily changing status, and why conventionally licensed constructions can still feel wrong under processing pressure. In that sense, (un)grammaticality can be illusory (Fanselow, 2021).

Distinguishing grammatical status from subjective ratings. Grammaticality research has to distinguish grammatical status, the conventional, population-level property made precise in §3.3, from the subjective read-outs that speakers provide. Those read-outs have two channels in the formal model: anomaly and confidence. (I use "objective" below only in that sense, population-level and convention-constituted, not judge-independent.)

Table 2 summarizes the two levels of analysis.

Table 2: Grammatical status versus subjective read-out

Level	Nature	Observable through
Grammatical status	Posterior existence of a licensed assembly in c	Converging evidence: corpora, production, repair behaviour
Subjective read-out	Anomaly signal F and confidence Φ	Ratings, online measures, self-reports, test–retest, explicit confidence

The measurement problem is familiar from other sciences: temperature can't be directly observed but has to be inferred through the behaviour of thermometers. Grammaticality likewise has to be inferred from several measures, with acceptability ratings as one imperfect window. The same caution applies to language-model probabilities: raw string probability by itself doesn't separate grammatical from ungrammatical strings, and minimal-pair probability contrasts are informative only when the intended message is controlled (J. Hu et al., 2026).[3]

The formal counterpart of this status/read-out picture, including what it implies for interpreting the factor analyses that are common in experimental syntax, appears in the formal core (§3.7, §3.8):

[3]Recent real-patterns work on language models raises the broader question of where linguistic patterns reside–in outputs, cognition, model-internal compression, or a plurality of models (Nefdt & Ladyman, 2026). OVMG does not settle that question for language as a whole. Its narrower claim is that grammaticality, for the purposes tracked here, is a projectible pattern in situation-indexed licensing profiles.

because such analyses target the structure of the subjective read-out, not grammatical status directly, a factor counts as evidence about the grammar only when it also shows up in the independent indicators for licensing, compatibility, and confidence.

This also explains why satiation and gradient judgments don't automatically threaten categorical status. A construction can remain unlicensed in the community's system while triggering progressively weaker negative responses through familiarization. Conversely, subjective response can remain continuous even when the population licensing state is effectively categorical (§4.4). Treating ratings as evidence rather than definition keeps grammatical hypotheses empirically vulnerable without collapsing the linguistic system of a communicative situation into individual responses to it.

Misattribution effects. Sometimes the subjective read-out of ungrammaticality doesn't accurately reflect conventional grammatical status. Both false positives (grammatical constructions feeling ungrammatical) and false negatives (ungrammatical constructions escaping detection) occur systematically.

When grammatical constructions feel ungrammatical. Listeners and readers can misanalyze utterances, triggering feelings of ungrammaticality even though the construction conforms to standard patterns:

(20) *The old man the boats.* (Ritchie & Thompson, 1984)

On first reading, one might treat *old man* as a noun phrase and arrive at nonsense. But the sentence actually has *the old* as the subject (meaning 'the elderly'), and *man* as a verb. Interpreted correctly, the sentence means 'The elderly operate the boats', and is fully grammatical.

Processing overload provides another source of misattribution. Heavily nested structures like (15) are grammatically well-formed but trigger negative responses due to cognitive limitations rather than grammatical violations.

When ungrammatical constructions escape detection. Conversely, conventionally ungrammatical utterances may fail to trigger negative responses when the intended meaning is strong:

(21) * *In Michigan and Minnesota, more people found Mr. Bush's ads negative than they did Mr. Kerry's.* (Pullum, 2009)

The intended interpretation is clear, and hearers readily infer the meaningful comparison. Under the relevant syntactic analysis, the comparison composes differently. The first clause suggests comparing numbers of people, but the second clause has *they* ('more people'), making the meaning 'more people did x than more people did y', which is nonsensical. But this violation often escapes detection.

Agreement errors can similarly hide within complex noun phrases:

(22) * *The patchwork of laws governing background checks, addressing assault-weapons limits, and regulating open-carry practices help explain why people continue to be wounded and killed.* (modified from Philip B. Corbett's editing quiz Corbett, 2016)

The singular subject *patchwork* clashes with plural *help*, but the intervening plural nouns mask this violation. Cognitive resources devoted to processing complexity can prevent speakers from registering the agreement error.

2.5 Mechanisms of grammatical change

The stability of form–value relations isn't static. Various motivations alter the evidence balance for new and established forms, explaining how grammatical systems evolve over time. These aren't a loose catalogue of causes: each is a route by which the evidence balance of §4.3 can change sign, so each predicts a direction of change (which forms should gain licensing and which should lose it) and feeds the actuation dynamics of §4.8.

Semantic motivations for reanalysis. Speakers regularly deploy metaphors, analogies, and context-induced reinterpretations that stretch or shift the meaning of existing forms. Over time, such re-analyses can yield new grammatical constructions that weren't previously part of the language's repertoire.

A well-documented example is the development of the *going to* futurate construction. Historically, *going to* described physical motion toward a place:

(23) *I am going to London.*

Frequent usage in contexts where motion implied subsequent action (e.g. *I am going to fetch some firewood*) led to semantic shift. The directional component was reinterpreted as marking future intention rather than spatial goals:

(24) *It's going to rain.*

What once expressed physical trajectory now serves as a predictive futurate marker. The form–value relation changed through semantic reanalysis, driven by communicative utility.

Social motivations for licensing and excluding forms. Linguistic variants often serve as markers of regional background, class, ethnic identity, or age group. These social cues motivate speakers to favor some forms over others; over time, those preferences shift what counts as grammatical.

The decline of the simple past (*le passé simple*) in modern spoken French illustrates social motivation. Forms like *il alla* ('he went') became associated with formal, literary, or provincial speech. To avoid signaling undesirable social affiliations, speakers gravitated toward compound forms like *il est allé*, which lacked these status-laden connotations.

Situational licensing tracks changing social dynamics: forms that were once fully grammatical can become socially marked and gradually excluded from normative grammar.

Structural motivations. Structural considerations arise from cognitive and communicative demands, including processing limitations, the need for clarity, and the influence of analogy.

Processing constraints and memorability. Transmission tends to favor patterns that minimize processing demands. Forms requiring multiple long-distance dependencies or heavy embedding are less likely to achieve stable transmission across generations. This pressure toward processability shapes grammatical evolution, with simpler patterns spreading through communities.

Analogical extension. When speakers encounter new communicative challenges, they often extend known patterns to new contexts:

(25) ? *I asked about what did she do.*

Such forms are licensed in some English varieties and marginal or rejected in others. Where they spread, the analogical source is transparent: main-clause interrogatives such as *What did she do?*. If licensed in a new community, this analogical extension would represent structural motivation reshaping the grammar.

Iconic motivations. Forms are sometimes licensed precisely because their structure reflects their meaning. Reduplication for intensity provides a simple example:

(26) *a big big problem*

The formal repetition iconically mirrors the magnitude being expressed. Such patterns are readily interpretable and likely to spread because the form–value link is immediately apparent.

Iconicity remains syntactically and variety constrained, though. In the target standard-English frame here, adjectival reduplication is natural in pre-head modifiers but rejected as a predicative complement:

(27) * *the problem was big big*

These motivations (semantic, social, structural, and iconic) can interact and sometimes conflict. While Optimality Theory has productively modeled such competing pressures through ranked constraints (Prince & Smolensky, 2004), OVMG views their resolution as arising from usage within communicative situations rather than solely from fixed rankings. The specific weighting of motivations reflects historically established patterns and communicative needs.

3 A formal core: grammaticality as conditioned stability

The framework separates two explanatory targets. A state theory specifies the conditions under which a form–value pair is grammatical *for a population in a communicative situation* at time t; a dynamics module explains how such states arise, persist, and change.

In projectibility terms, the state theory specifies the profile, while the dynamics module states its support: what stabilizes the profile and, where the evidence warrants, what maintains it. Support claims come in grades. Stabilization, maintenance by a production–repair loop, and control by a deviation-detecting process are distinct claims, and none follows from the framework's name (Reynolds, 2026a, 2026b). In Woodward's interventionist terms, a stabilizer earns that status when an intervention on it changes the profile in an invariant, predictable way (Woodward, 2001, 2003).

3.1 Conditioning structure

Let $c \in \mathcal{C}$ be a conditioning state, a construed communicative situation in Wiese's (2023) sense, together with whatever norm-centre is treated as relevant. Nothing here requires c to be externally given; interlocutors can misalign about c, and c can be learned, split, and renegotiated over time.

For many purposes it helps to think of c as a bundle that may include situational features (activity type, medium, footing), stable baselines tied to speaker ascription, and norm-orientation (discourse-community identification), but the formalism treats c abstractly as the conditioning variable that selects the relevant distribution of form–value relations.

3.2 Objects

Notation is typed as follows. In §3, f is a form, v is a value, and u is used only when the value is recoverable from context. In §4, a niche candidate is written $x \in \mathcal{V}^{\star}_{n,t}$ and abbreviates a candidate form–value realization. Constructional nodes are κ, assemblies are A, speaker-level inclusion states are $z_{j,t}$, and population licensing rates are θ_t.

Expository role	Formal object (§3)	Work in the formal core
coverage / structural viability	$1\left[\mathcal{A}(f,v) \neq \emptyset\right]$	structural viability: some covering assembly exists
operator compatibility	$\mathrm{def}(A)$: unification	hard operator-value compatibility
content oddness	plausibility term in R_i (§3.7)	content-oddness as a subjective cost
situational licensing	Beta posterior over θ_t; $L_t(A,c)$	situational licensing as mean *and* concentration
status profile	G_t^θ existence variable; $G_t = \mathbb{E}[G_t^\theta]$	graded status and uncertainty; simple products are limiting cases (§3.3.1)
decision threshold	$\tau(c)$, §3.3.4	decision-theoretic read-out only

Table 3: Where the expository roles from the opening sections live in the formal core. The formal implementation uses assemblies, unification, licensing posteriors, and decision read-outs.

The core's unit is the construction, in the CxG sense of a learned form–value pairing at any grain (Fillmore et al., 1988; Goldberg, 1995). Formally, a CONSTRUCTION κ is a pair (T_κ, μ_κ): a form template with typed slots, and a value contribution mapping slot fillers to content together with a PARTIAL OPERATOR ASSIGNMENT ω_κ. Nothing here is level-specific: a construction can be the plural cell of a lexeme, a clause pattern, an intonational contour, or a code-mixed frame.

Operator assignments live in a space Ω of partial functions from a dimension set $O(c)$ to finite value sets. The dimensions include clause type, polarity, tense–aspect anchoring, role allocation, scope, reference tracking, and information-packaging instructions. Which dimensions belong to $O(c)$ is itself conventional: a contrast enters only when the community has conventionalized it as an operator contrast in the sense of §2, so the inventory is community-indexed. Two partial assignments UNIFY, written $\omega \sqcup \omega'$, iff they agree on every dimension where both are defined; otherwise the join is undefined.

An ASSEMBLY A is a finite tree of constructions with all slots filled. Its yield is $\mathrm{form}(A)$; its operator assignment is $\omega(A) = \bigsqcup_{\kappa \in A} \omega_\kappa$, defined iff all contributions unify; its value is $\mathrm{val}(A)$. Three predicates over assemblies do the core work:

$$\mathrm{def}(A) = 1\left[\omega(A) \text{ is defined}\right] \qquad \text{(compatibility)} \qquad (28)$$

$$\mathrm{sat}(A,c) = 1\left[\omega(A) \text{ is total on } \mathrm{OBL}(c,A)\right] \qquad \text{(saturation)} \qquad (29)$$

$$L_t(A,c) = \textstyle\bigwedge_{\kappa \in A} Z_t(\kappa,c) \qquad \text{(licensing)} \qquad (30)$$

Compatibility is hard: there's no weighting an operator clash away. Saturation refers to the obligatory dimensions $\mathrm{OBL}(c,A)$, which aren't stipulated per language but derived from the licensing state. A dimension is obligatory exactly when zero-marking assemblies have been preempted to near-zero licensing in every niche where it's at issue (§4.5). Thus sat is a derived macro over licensing, not an independent primitive. For an assembly A of frame type $\varphi(A)$, $\mathrm{OBL}(c,A)$ abbreviates $\mathrm{OBL}(c,\varphi(A))$. English progressive marking and Turkish evidential marking are the worked cases there.

Licensing is latent, and the Beta is typed over a rate rather than a fixed binary fact. Let $z_{j,t}(\kappa,c) \in$

$\{0, 1\}$ be speaker j's local inclusion state for construction κ in c. The population state is the exchangeable licensing rate

$$\theta_t(\kappa, c) = P_j(z_{j,t}(\kappa, c) = 1)\,.$$

For a randomly sampled exchangeable speaker, $Z_t(\kappa, c) \mid \theta_t(\kappa, c) \sim \text{Bernoulli}(\theta_t(\kappa, c))$ is the predictive inclusion state for that speaker, and $L_t(A, c)$ is the event that the sampled speaker licenses all nodes in A. The definition needs a joint predictive distribution over the vector $\{Z_t(\kappa, c) : \kappa \in A\}$. The default approximation is within-speaker node-independence given $\boldsymbol{\theta}_t$, with dependence among nodes carried by the posterior over $\boldsymbol{\theta}_t$. That approximation should degrade where inclusion states cluster into lects, the heterogeneous region below; a richer model could replace it with an explicit within-speaker joint.

Nobody observes θ directly. Under exchangeability, de Finetti's representation theorem supplies a mixing measure over θ (de Finetti, 1937). The Beta family is the conjugate specialization that adds linear predictive updating: in the Bernoulli case, linearity of the posterior expectation of success in the counts is the binary analogue of Johnson's sufficientness postulate in Zabell's generalization (Zabell, 1982). Its mean is $C_t(\kappa, c)$ and its concentration is $\nu_t(\kappa, c)$ as in §4.3; the linear-in-counts update rule and the Beta prior family are one modelling assumption rather than two. The vector $\Lambda_{i,t}$ collects speaker i's inclusion states $z_{i,t}(\kappa, c)$, and the population distribution over those vectors supplies the population-level status state.

Compositional inheritance matters here. A novel token such as *She texted him the address* needn't have been encountered: the assemblies covering it are built from nodes–ditransitive transfer, pronominal object order, the relevant lexical-category relations–each with its own posterior, and the token inherits its status through them. What the learner updates is constructional regions, not memorized sentence types, so first-heard sentences aren't predicted to have zero licensing. Actual statistical pooling would require hierarchical hyperpriors over construction families, lexemes, and situations; this core needs only the compositional inheritance just described.

3.3 Grammatical status as a conditioned state property

3.3.1 Status as posterior existence

For a form f, value v, and conditioning state c, let $\mathcal{A}(f, v)$ be the set of assemblies with yield f and value v. The theta-conditional existence probability is

$$G_t^{\theta}(f, v, c) = P\Big(\, \exists A \in \mathcal{A}(f, v) :\ \text{def}(A) \wedge \text{sat}(A, c) \wedge L_t(A, c) \;\Big|\; \boldsymbol{\theta}_t, c\Big), \tag{31}$$

a random variable through the posterior over $\boldsymbol{\theta}_t$. Grammatical status is its posterior mean:

$$G_t(f, v, c) = \mathbb{E}_{p(\boldsymbol{\theta}_t \mid \mathcal{D}_t)}\left[G_t^{\theta}(f, v, c)\right], \tag{32}$$

where $\mathcal{D}_t$ is the evidence to date: tokens, structured omissions, repairs, avoidance, and situation cues. Status is a property of a population–situation pair, read through a learner's or an analyst's posterior; the population state it estimates is the distribution of node posteriors induced by c. Where value is recoverable from context, I write $G_t(u, c)$ and $G_t^{\theta}(u, c)$ for readability. For a single pivotal node, $G_t^{\theta} = \theta_t(\kappa, c)$, so the mean and concentration are literally the Beta posterior's mean and concentration.

Two scope conditions matter. First, for a finite yield and a finite, non-empty-recursive construction inventory (no cycles of null-yield expansions–the standard offline-parsability condition), $\mathcal{A}(f, v)$ is finite, so the existential is a finite disjunction. No additional tractability stipulation is needed.

Second, the relation to simpler heuristics. In the high-precision limit, where every node posterior has concentrated, the pair (31)–(32) has the same limiting classification as a weakest-link rule over the best assembly: a licensed, defined, saturated assembly exists or it doesn't. And under within-speaker node-independence, posterior independence across nodes, and a single dominant assembly A^*, the mean factorizes,

$$G_t(f, v, c) \approx 1\left[\text{def}(A^*)\right] \cdot 1\left[\text{sat}(A^*, c)\right] \cdot \prod_{\kappa \in A^*} C_t(\kappa, c). \tag{33}$$

A three-factor product heuristic can work in easy cases as the single-assembly, independent, high-precision special case of (31)–(32), with structural viability as non-emptiness and hard compatibility folded into def. Away from that limit, (31) supplies the existence variable and (32) its posterior mean, so uncertainty remains part of the status profile.

A three-factor heuristic can also hide content-oddness inside compatibility–the concentration of construal for cases like *colorless green ideas*. That work lives instead in the subjective read-out's cost vector (§3.7). A lexically bizarre but operator-compatible construal does not lower G_t and shows up as a plausibility cost in R_i: the sentence is grammatical and odd, matching the target contrast.

3.3.2 Projectible use of the posterior state

Grammaticality projects over this posterior state. G_t is the mean of a posterior random variable whose dispersion remains available for projection, so Figure 2 depicts the epistemic status state rather than a commentary on a scalar that discards half of it.

Some generalizations run over the posterior mean. Production and repair behaviour at $t + 1$ depend partly on G_t:

$$\Pr(\text{produce } u \mid n, c, t) \propto G_t(u, c)\ \rho_t^\star(u \mid n, c), \tag{34}$$

$$\Pr(\text{repair } u \mid c, t) = r_0 + r_1\big(1 - G_t(u, c)\big)\, r(\Delta, \iota), \qquad r_1 > 0. \tag{35}$$

The repair expression is the Δ-sensitive form used in §4.3; if Δ and footing are marginalized out, it reduces to a linear function of $1 - G_t$. This scopes the controller claim: transparent licensing failures with low update divergence should attract only baseline understanding-repair and be maintained mainly by preemption, while operator mis-settings draw the Δ-scaled repair flow; correction rates should instead track ascription and prescriptive dissonance.

Other generalizations run over epistemic concentration, and they're the diagnostic ones. Two form–value pairs can share a low mean while differing in how much evidence backs it, and the update dynamics make them behave differently under new exposure: a diffuse posterior moves with framing and familiarization; a concentrated one doesn't. This epistemic dispersion is $\text{Var}(G_t^\theta \mid \mathcal{D}_t)$, or equivalently the pivotal node's concentration in single-node cases.

Population heterogeneity is different. A community can have a sharply estimated interior θ because speakers are genuinely split, as with *might could* in a mixed community. That state has low epistemic variance but high between-speaker disagreement. The interior estimate counts as genuine heterogeneity only when no independently specifiable refinement of c (§3.6) splits the population toward the attractors; if such a partition exists, the apparent heterogeneity was a mis-specified conditioning state. Within-speaker test–retest instability and framing lability identify epistemic uncertainty; between-speaker disagreement with within-speaker stability identifies heterogeneity. Satiation and framing rank order turn on $1/\nu$, while moribund-contrast dispersion is measured between speakers because bounded memory lets individual histories diverge.

The status regions are:

- licensed: $G_t \approx 1$, $\mathrm{Var}(G_t^\theta \mid \mathcal{D}_t) \approx 0$;
- excluded: $G_t \approx 0$, $\mathrm{Var}(G_t^\theta \mid \mathcal{D}_t) \approx 0$–confident rejection, whether by structural failure or by concentrated non-licensing;
- unsettled: $\mathrm{Var}(G_t^\theta \mid \mathcal{D}_t)$ high, in two subtypes–*starved* (low concentration from a sparse opportunity set, as with independent relative *whose*) and *avoidance-divided* (low concentration despite dense opportunities, from conflicting or absorbed evidence flows, as with defective cells, §4.6);
- heterogeneous: posterior concentration high but the estimated population rate is interior–stable within speakers, divided across speakers.

The projectible generalizations–satiation profiles, framing sensitivity, transmission fidelity, repair rates, actuation–run over a state that carries the dimension they turn on. Left-branch extraction and *whose* approach similar means and part on concentration; clean defective cells such as *pobedit'* 1SG part from preempted gaps the same way. The feedback loop is closed: production and repair generate the later evidence streams, individual behaviour updates population licensing, and the population state projects back into future production and repair.

Strictly, the object doing the projecting is the licensing profile (G_t, ν) plus any independently estimated population heterogeneity, not the bare predicate GRAMMATICAL. The classical predicate names the high-mean, high-concentration region of that profile. It projects only those expectations that don't turn on concentration or heterogeneity; it doesn't by itself predict processing cost, production frequency, language-model string probability, typological comparanda, or transfer across conditioning states without the situation posterior $q_h(c \mid e)$.

3.3.3 Categorical talk as posterior concentration

Communities often treat grammaticality as a membership fact: a relationship counts as an available resource in c or it doesn't. On this account, that talk is licensed exactly where the posterior has concentrated–where the licensing state sits near an attractor and $\mathrm{Var}(G_t^\theta \mid \mathcal{D}_t) \approx 0$. Categoricality is a property of the state, not a threshold imposed on it; §4.4 derives the bimodality that makes the categorical regime the typical one for closed, high-opportunity, update-configuring contrasts. In the contested middle, categorical talk is predicted to be unstable and stakes-sensitive.

3.3.4 A decision-theoretic read-out

The threshold belongs in the read-out. A judge in c chooses between IN-REPERTOIRE and NOT-IN-REPERTOIRE under asymmetric losses $L_{\mathrm{fa}}(c)$ and $L_{\mathrm{fr}}(c)$. The optimal rule accepts iff

$$G_t(u, c) \geq \tau(c) = \frac{L_{\mathrm{fa}}(c)}{L_{\mathrm{fa}}(c) + L_{\mathrm{fr}}(c)}. \tag{36}$$

$\tau(c)$ does no constitutive work: it's a decision criterion induced by payoff structure, strict in high-stakes institutional contexts, permissive in in-group ones. Where the posterior is bimodal, the

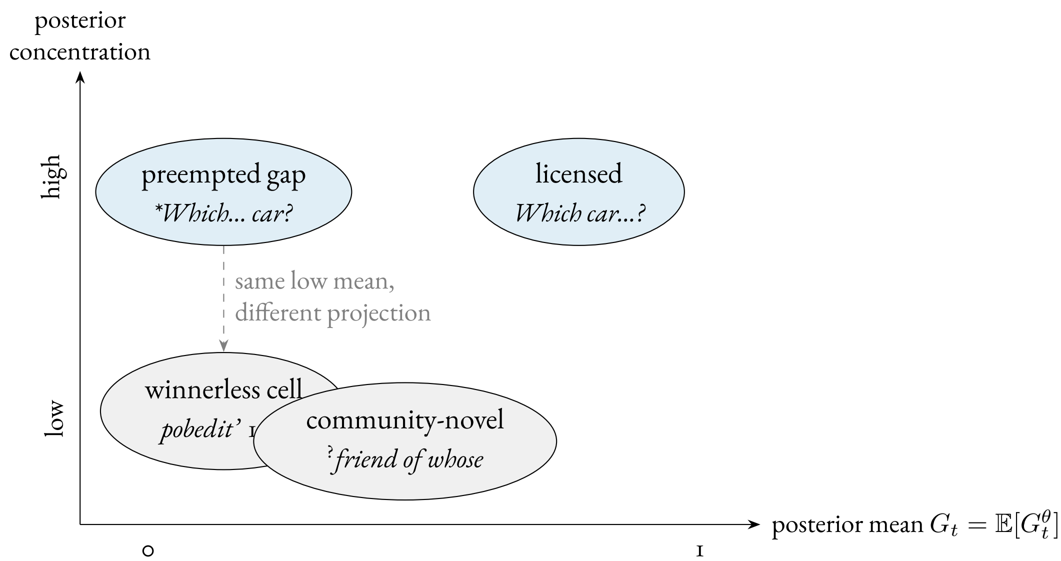

Figure 2: The epistemic status state has two dimensions. Preempted gaps (dense preemption, posterior concentrated near zero) and winnerless cells (transparent candidates, low mean but diffuse evidence) share a low mean and part on concentration, the axis that predicts satiation, framing sensitivity, and actuation. Stable between-speaker heterogeneity is a separate dimension, not shown.

read-out is insensitive to the exact value of $\tau(c)$; in the middle, judgments should be unstable and stakes-sensitive. A decision-layer discontinuity test can probe this read-out, but smoothness through $\tau(c)$ wouldn’t by itself refute the licensing-state theory.

Two clarifications keep the categorical read-out from overclaiming. First, calling the status “objective” means population-level and convention-constituted, not judge-independent in any stronger sense. The status stands apart from any individual’s response the way facts about word meaning or currency do: given the community’s practices, the projected relations aren’t made available by the analyst’s decision to group cases together (Reynolds, 2026a). Conventional status would be an equally accurate name.

Second, the threshold’s arbitrariness is confined by the dynamics. Where licensing posteriors are bimodal, no point estimate of $\tau(c)$ is needed; in the contested middle, $\tau(c)$ is set by independent stakes cues.

3.4 What the existential does and doesn’t do

Structural coverage is the non-emptiness of $\mathcal{A}(f, v)$: genuine analyzability failure is the case where no assembly covers the string at all. Conditional on a fixed inventory, $G_t^\theta = G_t = 0$ from an existential over the empty set. With inventory uncertainty, the value is zero up to whatever posterior mass remains on an unobserved stored construction. Empty coverage is the categorical failure mode generated by structural failure alone.

A condition belongs in a form template only when violating it makes the intended value unrecoverable or the exponent unidentifiable as an instance of the construction. If the intended value is transparent and the exponent is identifiable, the issue is licensing or compatibility, not coverage. This keeps template typing from becoming a way to relabel any concentrated licensing failure as structural failure.

Analyzability is relative to the constructional inventory, stored wholes included, not to productive composition alone. Syntactically anomalous idioms make the difference visible: *by and large*, *far be it from me*, and *come Monday* have no analysis by the productive syntax of English, but they're fully conventional (Fillmore et al., 1988). They're covered by their own stored constructions–coverage by some node, productive or memorized, is what analyzability means here. The cost is explicit: non-emptiness isn't pre-conventional. What it contributes is a different kind of failure (no covering assembly at all), not a convention-free one.

No additional primitive is needed. A string with no covering assembly gets $G_t = 0$ conditional on the inventory; a preempted gap gets $G_t \approx 0$ from a covering assembly licensed to zero. Both are excluded, and they differ in *why*–no assembly versus licensed-to-zero assembly–a difference with an empirical signature: the first fails under every construal and resists all framing once the relevant inventory is fixed; the second has a transparent construal, resists framing by concentration (§3.3.2), and is predicted to be re-openable only by the actuation route of §4.8. The constitutive core stays minimal without losing the distinction.

Because the existential is over value-matched assemblies, homophonous strings can split by construal. A form with one clashing parse and one compatible parse is grammatical on the compatible value and excluded on the clashing value; ambiguity doesn't let a failed assembly contaminate a successful one.

3.4.1 Assigning a failure

The result is a four-step diagnostic procedure, applied before consulting acceptability data and checkable against the independent indicators of §3.6 at each step.

Ask first whether any assembly covers u: if none does, the failure is structural (**Can the have running*). If a covering assembly exists, ask whether its operator contributions unify: if every covering assembly sets some dimension twice incompatibly, the failure is compatibility (**I've finished it yesterday*–two values on the reference-interval dimension; and the information-packaging clash in Cuneo and Goldberg's island example is the same formal object on a different dimension (Cuneo & Goldberg, 2022)).

If the contributions unify, ask whether the obligatory dimensions are saturated: if not, the failure is saturation (unmarked progressive in an ongoing-at-issue English niche). Only then does licensing arbitrate: a defined, saturated assembly whose pivotal node the population withholds is a licensing failure (**I have 25 years* for age–the assembly is compatible, the value transparent, and the age-predication node sits at $C_t \approx 0$ under preemption by the BE-frame).

Because the dimension inventory $O(c)$ is itself conventionalized, what counts as a compatibility failure in c depends on which contrasts the community has conventionalized. The steps are ordered, not independent–and each step's verdict is checkable against evidence other than the judgment it explains, which is what keeps the procedure from reproducing the circularity of a pure community-verdict account.

3.5 Interlocutor misalignment about conditioning

Because c is construed, interlocutors can disagree about which conditioning state is in force. Let a hearer h maintain a posterior $q_h(c \mid e)$ over conditioning states given cues e (situational cues, ascription cues, stance cues, etc.). Then the hearer's expected status is

$$\hat{G}_t^{(h)}(u \mid e) = \sum_{c \in \mathcal{C}} G_t(u, c)\, q_h(c \mid e).$$

3.6 Identifying the conditioning state before prediction

The conditioning state c isn't a free parameter to be selected after a judgment is observed. For empirical work, c has to be fixed before predicting ratings or repair behaviour. The posterior $q_h(c \mid e)$ is the identification device, estimated from non-judgment cues such as activity type, medium, participant roles, speaker ascription, register cues, and overt norm orientation. A form like *might could* isn't rescued by choosing a sympathetic norm-centre after the fact; the model predicts different judgments only when the independent cues make that norm-centre probable.

This also blocks a competence/performance redux. C_t, the population licensing state, is identified by corpus-rate-per-opportunity, elicited production, and repair behaviour, not by acceptability ratings alone. F, the anomaly signal, is identified by ratings, online measures, and self-reports. The confidence read-out Φ (§3.7) is identified by test–retest stability and explicit confidence ratings, neither of which enters the estimation of the licensing posterior itself. Framing lability is a confirmatory outcome, not an identifying input. The model is falsified where these dissociate in the wrong direction: if satiation changes later corpus rates for a purportedly preempted gap, if repair behaviour tracks prescriptive dissonance rather than licensing, or if high-opportunity preempted forms behave like low-evidence uncertainties, the state assignment is wrong.

Reassignment is constrained. Localizing a failed prediction to a narrower conditioning state is warranted only when that state is specifiable independently of the surviving cases, by activity type, medium, participant roles, or norm-orientation fixed before the failures were seen (Reynolds, 2026a). A partition of $\mathcal{C}$ redrawn around exactly the successes redescribes failure as success; it doesn't rescue the model.

Confirmatory studies should register the conditioning partition and the protocol for estimating $\rho_t^\star$ (e.g. forced-choice norming over the competitor set before the licensing data is touched). The corpus checks in this paper are illustrative probes, not registered tests. Under that discipline, repeated post-hoc reassignment across future studies counts against the framework itself, not just against particular state assignments.

Two identification points follow from the fact that a corpus rate estimates the product of licensing and counterfactual choice (§4.2), not licensing alone. First, $\rho_t^\star$ has to be estimated from evidence that doesn't include the licensing data it later interprets. A forced-choice norming task supplies it: present the niche n and the competitor set $\mathcal{V}_{n,t}^\star$ to a separate sample and record how often each candidate is chosen when stipulated to be available. That yields $\rho_t^\star$ before any corpus rate for u is consulted, so a low rate can be diagnosed as non-licensing ($C_t \approx 0$, $\rho_t^\star$ high) rather than licensed-but-dispreferred selection (C_t high, $\rho_t^\star \approx 0$).

Second, $\tau(c)$ isn't fit to the judgments it predicts. Where the licensing posterior is bimodal (§4.4), G_t is insensitive to the exact threshold, so no point estimate of $\tau(c)$ is needed; in the contested middle, $\tau(c)$ is set by independent stakes cues (institutional versus in-group footing), and the movable-cutoff prediction of §3.3.4 tests the decision layer directly: manipulate the stakes and the location of any response-type switch should move, which can't happen if $\tau(c)$ were merely a curve fitted to the responses.

The niche n is disciplined in the same way as c. Before an absence is counted as preemption, the analyst has to specify the communicative job, the recoverable intended value, the competitor set, and the outside option without using the target's acceptability as the criterion. The counterfactual choice term $\rho_t^\star(u \mid n, c)$ is then normed independently, for example by forced choice under a stipulation that all candidates in the set are available.

A low-rate candidate counts as preempted only when $N_t(n, c)$ is large, $\rho_t^\star$ is non-negligible, and omissions are attributable to competitor choice rather than avoidance; otherwise the same absence is starved or divided. Classifications of low-mean cases should be reported with robustness checks across at least two plausible niche granularities.

3.7 The feeling of (un)grammaticality

Speakers register putative violations through a metacognitive signal that can diverge from status; the two are identified by different evidence streams (§3.6). The observable phenomenology is modeled as a pair: a signed anomaly signal and a confidence read-out. The pair is what lets the model derive, rather than assert, the contrast between confident rejection and ineffability.

Let $\hat{G}_{i,t}^{(h)}(u \mid e)$ be hearer i's expected status under situation uncertainty (§3.5). Write $R_i(u, e)$ and $I_i(u, e)$ for implementation costs and prescriptive dissonance after cue-based situation uncertainty has been integrated. The anomaly drive accumulates low expected status, implementation costs, and ideological overlays:

$$M_{i,t}(u, e) = \alpha_G\big(1 - \hat{G}_{i,t}^{(h)}(u \mid e)\big) + \gamma^\top R_i(u, e) + \delta_I\, I_i(u, e) + \varepsilon_i, \tag{37}$$

$$F_{i,t}(u, e) = -\frac{\max\{M_{i,t}, 0\}}{1 + \max\{M_{i,t}, 0\}} \in (-1, 0]. \tag{38}$$

The bounded link and the floor at zero encode the asymmetry of the signal: there's no positive feeling of grammaticality, only the negative signal registered on deviation. I_i is prescriptive dissonance. The cost vector R_i includes the processing components–the locality term $\mathcal{L}_{\text{loc}}(u)$ with its saturating form, interference, garden-path reanalysis, surprisal–and a PLAUSIBILITY component: the implausibility of the best construal given world knowledge and lexical expectations. That component handles content oddness without making it a status fact. *Colorless green ideas sleep furiously* is defined, saturated, and licensed node by node, so $G_t \approx 1$; its strangeness enters here, as a construal-plausibility cost, and the sentence comes out grammatical and odd, with no extra primitive.

The locality component can still be written explicitly:

$$\mathcal{L}_{\text{loc}}(u) = \sum_{d \in \mathcal{D}_{\text{dep}}(u)} \ell\big(|d|\big), \qquad \ell(k) = \begin{cases} k, & k \le K_{\text{sat}}, \\ K_{\text{sat}} + \beta_L\,(k - K_{\text{sat}})^{\eta_L}, & k > K_{\text{sat}}. \end{cases}$$

Here $|d|$ is the length of dependency d (in intervening discourse referents or words), K_{sat} is a saturation threshold beyond which additional distance contributes sublinearly, and β_L, η_L (with $0 < \eta_L < 1$) control the rate of that sublinear growth.

The second read-out is confidence. Let $\nu_i(u, e)$ be the concentration of i's posterior over the *pivotal condition*–the conjunct in (31) on which the token's status turns–after cue-based situation uncertainty has been integrated. Conditional on an independently fixed c, write $\nu_i(u, c)$; if the pivotal condition differs across high-probability conditioning states, the observable confidence read-out follows the modal condition and treats the remaining mixture as lower confidence. For structural and compatibility failures the pivotal condition is a fact about the assembly set, so once the relevant inventory is fixed that posterior is treated as effectively degenerate: confidence is high by modelling convention, not by a separate learning theorem. For licensing-pivotal tokens, ν_i is the Beta concentration of the pivotal node's posterior (§4.3). Define

$$\Phi_{i,t}(u,e) = \frac{\nu_i(u,e)}{\nu_i(u,e)+\nu_0} \in [0,1), \tag{39}$$

with ν_0 a scaling constant. The pair (F,Φ) is the observable phenomenology, and the predictions separate:

- *No assembly* (**Can the have running*): F strongly negative, $\Phi \approx 1$, under every construal.
- *Compatibility failure* (**I've finished it yesterday*): F strongly negative, $\Phi \approx 1$; entrenchment of the parts can't repair it, because unification is hard.
- *Preempted gap* (left-branch extraction if $\rho_t^\star$ norms high): F strongly negative, Φ high–"that's impossible"; framing-resistant.
- *Defective cell* (*pobedit'* 1SG; **amn't* as a borderline case): F negative, Φ low–"I don't know how to say it"; framing-movable when the outside option, rather than a winner, absorbs production.
- *Community-novel* (independent relative *whose*): F weakly negative, Φ low; framing-movable, with surprisal adding R-channel noise.
- *Processing overload* (centre embedding): F negative via R_i, Φ moderate; satiation lowers the R contribution, moving F toward zero without touching the licensing posterior.

Optionally, the model can carry an attribution variable: on anomaly detection the hearer infers a source $\zeta \in \{$UNLICENSED, OTHER-COMMUNITY, SPEAKER ERROR, NOISE, OWN OVERLOAD$\}$, with the posterior over ζ shaped by Φ and by the same ascription machinery as $q_h(c \mid e)$. That derives, rather than stipulates, why learner errors and cross-dialect tokens fail to trigger the same anomaly signal–they're attributed to OTHER-COMMUNITY or ERROR–and it gives the misattribution effects of §2.4 a formal home. The attribution layer is a measurement refinement, not part of status.

A direct falsifier follows: if measured confidence fails to dissociate preempted gaps from clean winnerless cells such as *pobedit'* 1SG at matched mean ratings, posterior concentration has no independent empirical role in the model.

3.8 Measurement: four observation channels

Status is latent; it's estimated from converging observation channels, none of which defines it. The observation model has four channels:

$P(\text{production} \mid \theta, n, c)$	choice among assemblies in a niche (§4.2)	(40)
$P(\text{repair} \mid \theta, \Delta, I, c)$	sensitive to operator mis-setting (§4.3)	(41)
$P(\text{judgment} \mid \hat{G}^{(h)}, R, I, \nu)$	the (F,Φ) read-out of §3.7	(42)
$P(c \mid \text{cues})$	situation inference, $q_h(c \mid e)$ (§3.5)	(43)

Production is choice among assemblies, not a direct read of licensing: a corpus rate estimates the product of licensing and counterfactual choice, exactly as in §4.2, and the identification discipline

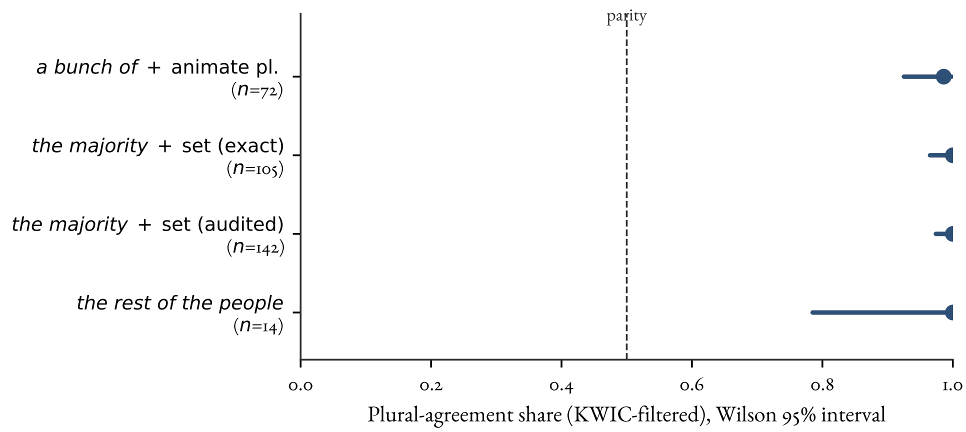

Figure 3: Corpus-rate-per-opportunity as an estimator of licensing for a prescriptively salient agreement choice. Plural-agreement share (dot) with Wilson 95% interval (bar) for collective-partitive subjects in COCA, after KWIC filtering to subject-position tokens; the dashed line marks parity (no directional preference). Data: `agr-coca-projection` probe (*bunch*, *majority*/*minority*, and partitive cells). Notional plural dominates in every cell, and the interval widens with the smaller denominators rather than drifting toward parity.

of §3.6 (norming $\rho_t^\star$ on independent data before the licensing data is touched) applies here. Repair enters with the sign predicted in §3.3.2 and with the Δ-sensitivity of the repair stream (§4.3). Judgment is a metacognitive read-out, not status; the confidence channel Φ is identified by test–retest stability and explicit confidence ratings, with framing lability reserved for confirmation. Situation is inferred from cues, not stipulated after the fact. A fitted version composes the channels as a joint likelihood over production, repair, judgment, and situation cues with shared latent $\boldsymbol{\theta}$, n, and c; identifiability requires at least one non-judgment channel for $\boldsymbol{\theta}$ and one non-repair estimate of Δ.

Factor analyses of acceptability ratings target the structure of the judgment channel; a factor counts as evidence about the grammar only when it also shows up in the independent indicators for licensing and compatibility. Judgment studies should report dispersion and test–retest structure alongside means, because the model's sharpest projections–satiation rank order, framing lability, the gap/cell contrast–run over the concentration that means discard.

The corpus-rate-per-opportunity indicator is concrete. Figure 3 shows one worked case: agreement on collective-partitive subjects (*a bunch of people*, *the majority*), where surface-head number (singular) and notional number (plural) diverge. The usage pattern is overwhelmingly one-sided, which makes the case useful as an audit trail. Counting only subject-position tokens after KWIC filtering (so non-subject false positives are removed) gives a plural-agreement share with a Wilson interval for each cell. The shares sit far above parity in every cell, with the interval width tracking the token count rather than the direction. The example shows how a licensing estimate can be read off usage per opportunity rather than ratings, with uncertainty made explicit.

Table 4 makes the figure auditable: exact filtered cell counts, so the estimate can be checked rather than taken on trust.

Table 4: Filtered agreement cells behind Figure 3. Counts are KWIC-filtered subject-position tokens in COCA; "pl." and "sg." are plural- and singular-agreement targets. The surface-head baseline predicts singular; the observed share is plural.

Cell	pl.	sg.	N	Plural share (Wilson 95%)
a bunch of + animate pl.	71	1	72	0.986 (0.925–0.998)
the majority + set (exact)	105	0	105	1.000 (0.965–1.000)
the majority + set (audited)	142	0	142	1.000 (0.974–1.000)
the rest of the people	14	0	14	1.000 (0.785–1.000)

3.9 Worked examples: deriving status from the posterior

The examples from §2.2 show how the posterior assigns status. The winnerless cell adds the case where concentration derives the phenomenology rather than merely labelling it.

1. **Can the have running* (nonsense). $\mathcal{A}(f, v) = \emptyset$: no covering assembly in the fixed inventory. $G_t^\theta = G_t = 0$ up to inventory uncertainty; $\Phi \approx 1$ under every construal. Status: ungrammatical, structurally.
2. *Colorless green ideas sleep furiously*. A covering assembly is defined, saturated, and licensed node by node: $G_t \approx 1$. The strangeness is a construal-plausibility cost in R_i: F negative, status grammatical. Content-odd, not ill-formed.
3. **I've finished it yesterday* (compatibility failure). Covering assemblies exist, but every one sets the reference-interval dimension twice incompatibly–the perfect's contribution and the deictic adverb's don't unify–so def$(A) = 0$ throughout. $G_t^\theta = G_t = 0$; $\Phi \approx 1$. Entrenchment of the component constructions can't repair it: the clash is outside the licensing terms altogether.
4. **I have 25 years* (licensing failure). Defined and saturated; the value is transparent. The pivotal age-predication node sits at $C_t \approx 0$ with high ν, preempted by the BE-frame in a dense niche. $G_t \approx 0$, Φ high. Status: ungrammatical, by concentrated non-licensing.
5. *?friend of whose* (community-novel). Defined and saturated; the niche is vanishingly sparse, so the pivotal node's posterior sits near its prior with low ν. G_t intermediate, $\mathrm{Var}(G_t^\theta \mid \mathcal{D}_t)$ high, Φ low. Status: unsettled (starved); framing-movable, with surprisal adding R-channel noise.
6. **Which did you buy car?* (putative preempted gap). Defined and saturated; the dedicated discontinuity node–the only activated node covering the determiner–head discontinuity–receives negative evidence from fronting opportunities only in proportion to independently normed $\rho_t^\star$. If $\rho_t^\star$ is high, its posterior is driven to zero and concentrates there (§4.3); if $\rho_t^\star$ is low, the case belongs closer to starvation. Status assignment requires the niche-norming protocol of §3.6.
7. *pobedit'* 1SG (winnerless cell; **amn't* as a mixed case). The niche is dense and candidate assemblies are transparent, but no candidate accumulates positive evidence–the outside option absorbs production, and because avoidance has low attributability, candidates accumulate little confident negative evidence either (§4.6). Every candidate's pivotal node has low mean and low ν. G_t low, $\mathrm{Var}(G_t^\theta \mid \mathcal{D}_t)$ high, Φ low. Status: unsettled (avoidance-divided); phenomenology ineffability, not rejection–same mean as a preempted gap, different concentration, different feeling.
8. *The bread the baker the apprentice helped made is delicious* (processing overload). G_t high; the locality term in R_i is large. F negative, Φ moderate; satiation and instruction move the R contribution, not the licensing posterior. Status: grammatical, transiently ill-felt.

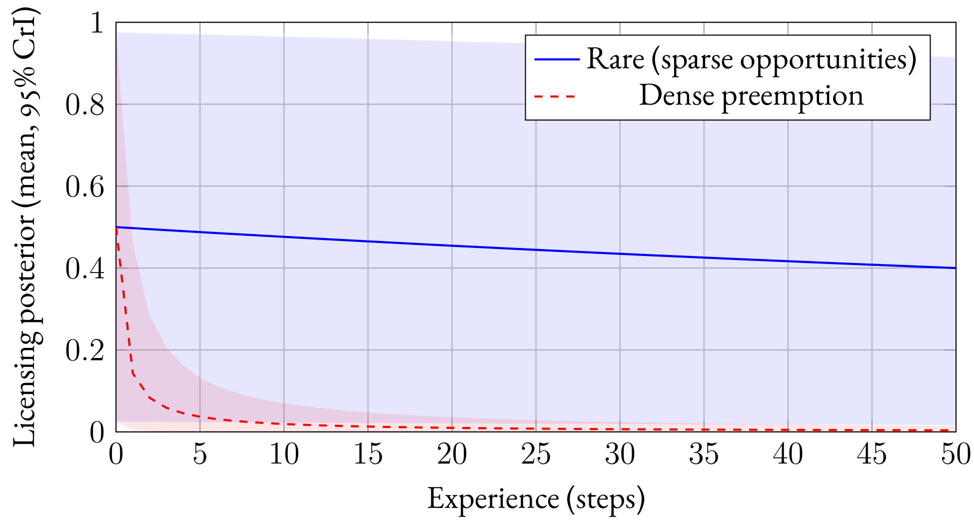

Figure 4: Licensing posteriors under preemption, from a Beta$(1, 1)$ prior with per-step effective preemption mass 0.01 (rare, blue) versus 5 (dense preemption, red); lines are posterior means, shaded regions 95% credible intervals. The rare construction stays near its prior with a credible interval spanning nearly the whole unit interval, while the dense-preemption posterior both falls and concentrates. The projections in §3.3.2 that turn on evidence structure–satiation, framing sensitivity, actuation–run over this dispersion difference, which means alone don't encode.

Figure 4 shows the low-opportunity versus dense-preemption contrast that underwrites several of these classifications.

4 Dynamics: why the state stabilizes

The dynamics module explains trajectories of construction-level licensing posteriors, and in principle refinements of the conditioning partition itself. It doesn't define grammaticality.

4.1 Niches, competitors, and opportunity sets

Let n index a constructional niche (a communicative job), and let $\mathcal{V}^{\star}_{n,t}$ be the opportunity set: the observed, licensed, and analogically generated candidate realizations that could plausibly do that job in some conditioning states. Let $N_t(n, c)$ be the number of opportunities for niche n in conditioning state c over some time window, and $k_t(x, n, c)$ the observed count of candidate realization $x \in \mathcal{V}^{\star}_{n,t}$.

The star matters. Preempted gaps and innovations often involve candidates with no positive tokens. Their counterfactual utilities are still defined because the hierarchy $\mathcal{H}$ generates analogical candidates. English learners can represent a left-branch extraction candidate by combining interrogative fronting, modifier–noun packaging, and question formation, even if no one licenses that candidate as an English resource. The same machinery lets a novel but ordinary sentence inherit coverage from licensed higher-grain constructions rather than being preempted merely because that exact sentence type is new.

4.2 Usage as licensing × choice among candidates

Separate LICENSING from SELECTION. A candidate can be licensed but rarely chosen.

Let $\rho_t^\star(x \mid n, c) \in [0,1]$ be the counterfactual probability of choosing x if it were available in niche n under c. A flexible choice model is a softmax over utilities:

$$\rho_t^\star(x \mid n, c) = \frac{e^{U_t(x;n,c)}}{\sum_{x' \in \mathcal{V}_{n,t}^\star} e^{U_t(x';n,c)}}, \qquad U_t(x;n,c) = \mathrm{w}^\top \mathrm{f}(x;n,c).$$

At the population level, the expected usage rate is $\pi_t(x \mid n, c) \approx G_t(x,c) \cdot \rho_t^\star(x \mid n, c)$. In single-pivotal-node cases this reduces to $C_t(\kappa_x, c) \cdot \rho_t^\star(x \mid n, c)$, where κ_x is the node that would realize x in c. The counterfactual choice term can be high even when $G_t(x,c)$, or the pivotal node's C_t, is near zero; absence becomes informative under that condition.

4.3 Dynamics: evidence streams, weighted omissions, and repair

The learner represents each construction–situation pair (κ, c) with a Beta posterior over the population licensing rate $\theta_t(\kappa, c)$ (§3.2). With perfect retention, the update is the ordinary conjugate filter for exchangeable Bernoulli licensing observations. With memory discount $\delta_m \in (0,1]$, an evidence item at occasion t_i contributes $\delta_m^{t-t_i}$ of its weight at time t. Setting $\delta_m = 1$ recovers the fixed-parameter posterior; $\delta_m < 1$ is a power-discounted dynamic filter for a drifting convention (West & Harrison, 1997), which is what the destabilization result in §4.7 turns on.

For a target candidate realization x, the streams are positive evidence s_t, error evidence e_t, and preemption mass p_t, distributed over activated nodes by stream-specific evidence weights:

$$a_{\kappa,c,t+1} = \delta_m a_{\kappa,c,t} + \lambda_\kappa^+(x,c) s_t(x,c), \tag{44}$$

$$b_{\kappa,c,t+1} = \delta_m b_{\kappa,c,t} + \lambda_\kappa^-(x,c)\{e_t(x,c) + p_t(x,c)\}. \tag{45}$$

The node-level mean and concentration are

$$C_{t+1}(\kappa, c) = \frac{a_{\kappa,c,t+1}}{a_{\kappa,c,t+1} + b_{\kappa,c,t+1}}, \qquad \nu_{t+1}(\kappa, c) = a_{\kappa,c,t+1} + b_{\kappa,c,t+1}. \tag{46}$$

The two weights encode the grain asymmetry used below. Positive evidence spreads according to the parsing posterior,

$$\lambda_\kappa^+ = P(\kappa \in A^* \mid x, c, \mathcal{D}_t),$$

so a heard token confirms the nodes that plausibly cover it. Error and omission evidence attaches by likelihood sensitivity,

$$\lambda_\kappa^- \propto \left| \frac{\partial \log P(\mathcal{D}_t \mid \theta_t(\kappa, c))}{\partial \theta_t(\kappa, c)} \right|,$$

over activated nodes, so negative mass concentrates where the data would have looked different if that node were licensed.

Ascription weighting applies to all streams. Each token's contribution is scaled by the hearer's posterior that it's a competent realization of the norms of c. That's why child and learner errors barely move adult posteriors, and why koinéization falls out as a shift in ascription weights rather than a separate mechanism.

Weighted omissions. A simple preemption rule treats absence as evidential in proportion to the counterfactual choice probability: $p_t = \sum_o \rho_t^\star(x \mid n(x), c)$ over niche opportunities o. That approximation handles ordinary competitor choices but not occasions where the niche arose and nothing in the candidate set was produced–periphrasis, avoidance, repair-in-progress. Counting those as negative evidence would make every defective cell collapse into a preempted gap. The weighting follows from inverting the production model.

Consider an occasion i where niche n arose and some $y_i \neq x$ was produced. The evidential question is how strongly that choice tells against x's membership in the repertoire. The likelihood has to include the same production gate as §3.3.2. Let $g_i(x, e_i)$ be the producer-side expected status: the producer's expectation that x is available under the cues at occasion i, or the analyst's estimate of that expectation from observable cues. If x is licensed, its effective choice mass is $g_i(x, e_i)\rho_t^\star(x \mid n, c)$; if x is unlicensed, that mass is zero and the alternatives renormalize. The per-occasion evidence weight for non-licensing is

$$q_i = \left[1 - \frac{P(y_i \mid Z_x = 1, g_i, n, c)}{P(y_i \mid Z_x = 0, g_i, n, c)}\right]_+, \tag{47}$$

the attributability of the omission, where Z_x abbreviates availability of a licensed route for x and $[r]_+ = \max\{r, 0\}$. Negative values are evidence for x, not omission evidence against it.

Two regimes matter. When y_i is an in-repertoire competitor, the renormalization gives, under high confidence,

$$q_i = \rho_t^\star(x \mid n, c) \quad \text{when } g_i \approx 1.$$

The simple preemption mass is recovered for ordinary competitor choices. More generally, for small candidate mass,

$$q_i \approx g_i(x, e_i)\rho_t^\star(x \mid n, c).$$

Near a concentrated zero state, the gate is near zero, so omissions become almost uninformative about whether the candidate is licensed but never produced or unlicensed. Endogenous omissions maintain the attractor but no longer distinguish unlicensed candidates from licensed candidates that go unused, so re-opening a concentrated gap requires the actuation routes of §4.8.

When y_i is the outside option–periphrasis or avoidance, available at fixed utility under both hypotheses–its choice probability is nearly the same whether x is licensed or not, provided x's counterfactual utility is low or the production rule assigns a face cost to weakly licensed forms. The condition is independently estimable from the same production data that estimates $\rho_t^\star$: speakers who avoid an exposed candidate despite understanding the intended value are treating production itself as costly. Then $q_i \approx 0$: taking the outside option is nearly uninformative about x.

A second weight scales for niche identification: r_i is the posterior probability that occasion i instantiated a job x would have done at all, with the intended value recoverable under x. The generalized preemption mass is then

$$p_t(x, c) = \sum_{i:\,\text{omissions}} r_i q_i\, \delta_m^{t-t_i}, \tag{48}$$

which reduces to the simple $N_t \cdot \rho_t^\star$ rule when every omission is a high-confidence competitor choice in a clearly identified niche with $\delta_m = 1$. Neither weight is a free parameter: both are read off the production model and the niche posterior. That weighting keeps periphrasis from counting as confident negative evidence, separating the defective cell's low-mean/low-concentration state from the preempted gap's low-mean/high-concentration one (§4.6, §3.9).

The repair stream: a candidate controller. The paper's support claims come in grades: stabilization, maintenance, control (§5). The dynamics so far deliver the first two. The third needs a process that registers departures from the licensed profile and pushes the population back–and conversational repair is the obvious candidate, with a formal home.

Let $\Delta(d)$ be the expected common-ground divergence from mis-setting operator dimension d: the quantitative face of "update-configuring" in the working test of §2. Formally, $\Delta(d)$ is an expected KL divergence between the common-ground posterior induced by the intended value and the posterior induced by the mis-set value, in the same spirit as RSA-style update models (N. D. Goodman & Frank, 2016). When a token mis-sets d, other-initiated repair occurs with probability $r(\Delta, \iota)$, increasing in Δ and modulated by footing ι. Each repair event feeds the evidence streams: error evidence e_t for the deviant node, positive evidence s_t for the repaired-to competitor. Over a window with $N_t(n, c)$ opportunities, the repair-generated flow against a deviant convention in niche n is

$$\psi_{\text{rep}}(n) = N_t(n, c) \cdot P(\text{mis-set}) \cdot r(\Delta, \iota), \tag{49}$$

whose magnitude scales with the product of opportunity and divergence. This scalar is the mean-field contribution generated by repair-weighted positive and error evidence, not a fourth evidence stream in addition to s_t and e_t.

In the mean-field dynamics (§4.4) repair acts as majority enforcement: it increases the evidence flow toward whichever candidate is already modal in the relevant community state. This gives Woodward's interventionist signature: the repair rate earns controller status if an intervention on it changes the profile's return-to-attractor behaviour in an invariant, predictable way (Woodward, 2001, 2003). The claim stays graded: what the term establishes is a testable candidate controller, not a demonstration that grammar is controlled.

Two commitments come with the term. First, the operator/style boundary becomes a parameter statement: policing intensity is predicted to scale with $N_t \cdot \Delta$, with opportunity estimable from opportunity-annotated corpora and Δ from comprehension probes or annotated update-divergence tasks that ask what common-ground change the mis-set value would induce. The §2 falsifier–a closed, high-opportunity, non-configuring contrast policed like grammar–has a quantitative form: such a contrast has $\Delta \approx 0$, so its corrective flow collapses to a social-sanction baseline and its licensing distribution should stay gradient, policed like style. Second, if repair rates track social indexing rather than Δ, the corrective term collapses into sociolinguistic policing and the controller claim fails cleanly. Blythe and Croft's utterance-selection dynamics remain the $r \equiv 0$ special case, so the import is conservative (Blythe & Croft, 2012).

Grain asymmetry and compositional inheritance. The two kinds of evidence attach at different grains. Positive evidence distributes over activated nodes by the coverage weights: a heard token confirms every construction that covers it. Error and preemption mass instead concentrate on the most specific node that individuates the candidate within its niche, because that node is what distinctively predicts the missing tokens. In Bayesian terms, the likelihood penalty for structured non-occurrence falls on the hypothesis that wastes probability mass on sentence types never observed; likelihood is a quantitative measure of the weight of implicit negative evidence (Perfors et al., 2011).

Compositional inheritance raises an objection. Token-level licensing is assembled from constructional nodes, so left-branch extraction activates the well-licensed nodes for interrogative fronting, question formation, and modifier–noun packaging. The sum can look as though it should hand the candidate substantial licensing.

The coverage weights block that result. A node contributes to a token's pooled licensing only for the properties it actually sanctions, and those parent nodes sanction only *contiguous* realizations. None of them covers the determiner–head discontinuity that defines **Which did you buy car?*; producing that token requires a node that licenses a determiner–head arc crossing the verb, and no contiguity-preserving node in $\mathcal{H}$ generates one (Reynolds, 2026d).

The parents keep their high licensing, each confirmed by its own abundant contiguous tokens, and contribute none of it to the discontinuous token through the pooling sum. The dedicated left-branch-extraction node, the only activated node that covers the discontinuity, carries near-gating weight for it. It absorbs preemption mass only to the extent that the missed opportunities are attributable to competitor choice, which is why the independent $\rho_t^\star$ norming required in §3.6 is necessary for the assignment. The dedicated node is required by the derivation, not gerrymandered to force the result.

This asymmetry also answers an acquisition worry. If token-level licensing were just pooled parent licensing, learners should pass through a stage of treating left-branch extraction as live. They don't need to: the candidate is generated by analogy, so it's on the table with non-zero prior support, but its dedicated node has no scaffolding of its own, fronting opportunities are dense in the input, and each one adds preemption mass.

A companion corpus study quantifies one side of the evidential situation: across 1,228 fronting opportunities in a sixteen-corpus Universal Dependencies sample, zero genuine determiner–head discontinuities occur (95% upper bound 0.24%), while contiguity-preserving alternatives (pied-piping, fused-head construals, and the *big mess* family) occur at measurable rates (Reynolds, 2026d). Comprehension pushes the same way: a fronted *whose* or *which* defaults to a fused-head parse, so the discontinuous analysis also loses the parsing race (Culicover et al., 2022). What remains to be normed is $\rho_t^\star$: if speakers would rarely choose the discontinuous form even under a permissive stipulation that English allowed it, the absence would support starvation rather than preemption. The predicted non-satiation of left-branch extraction remains a test; Lu et al.'s (2024) meta-analysis of satiation in extraction from islands supplies the closest current analogue, not a direct measurement of LBE itself.

OPEN-LOOP LEARNING AND CLOSED-LOOP DYNAMICS. The discrete update (46) is the primary dynamics. With perfect retention ($\delta_m = 1$) and a fixed environment (exogenous evidence streams), it's the ordinary decaying-gain estimator: the posterior mean follows a hyperbolic trajectory of the form $a_0/(a_0 + b_0 + \bar{e}t)$, as plotted in Figure 4, and converges without bifurcating. With bounded memory ($\delta_m < 1$), it instead behaves as the stationary filter described in §4.4 and §4.7.

The normal-form approximation arises when the loop in §3.3.2 is closed: positive evidence is generated by production at a rate that grows with the current licensing state, while error, preemption, and repair-weighted evidence respond to what the population already does. Replacing those endogenous evidence streams by their expectations yields the cubic drift in (50). I use that ODE for qualitative fixed-point and comparative-statics arguments; the attractors belong to the closed-loop population dynamics, not to individual open-loop learning.

4.4 EMERGENT CATEGORICALITY AS BIMODALITY

The sense of EMERGENCE used here is weak and mechanistic. Population-level categoricality is derived from learner-level updating, selection, and repair; it isn't a label for whatever remains unexplained. This differs from Hopper's "emergent grammar", where grammar is perpetually provisional (Hopper, 1987). The model allows conditioned states with near-absorbing attractors.

In the mean-field approximation, the closed-loop dynamics have the normal form

$$\dot{C} = C(1-C)\big[\alpha C - \beta + \chi_{\text{rep}}\psi_{\text{rep}}(2C-1)\big], \tag{50}$$

for a focal candidate in a two-way niche. The boundary terms say that no candidate is reinforced when it's never produced or already saturated. The bracketed term is the expected evidence balance: αC is the positive feedback from production, β collects preemption, error, face cost, and outside-option pressure, and the repair term is majority enforcement. Repair pushes toward the current modal candidate, not toward an externally stipulated convention. When $0 < C^{\ddagger} < 1$, with

$$C^{\ddagger} = \frac{\beta + \chi_{\text{rep}}\psi_{\text{rep}}}{\alpha + 2\chi_{\text{rep}}\psi_{\text{rep}}},$$

the endpoints are attracting and $C^{\ddagger}$ is the separatrix. Across construction nodes, a community whose evidence rates are bounded away from zero should develop a bimodal distribution of licensing states: most nodes cluster near attractors, and only low-evidence, contested, or outside-option-dominated nodes remain diffuse.

The convergence claim has to match the update actually used here. With perfect retention ($\delta_m = 1$), the occasion-level system is close to reinforcement processes of Pólya type, where stochastic approximation for urn-type models supplies the relevant convergence technology (Pemantle, 2007). The signaling-game case, structurally the closest analogue, is treated directly by Argiento, Pemantle, Skyrms, and Volkov (Argiento et al., 2009), with an extension by Hu, Skyrms, and Tarrès (Y. Hu et al., 2011).

The paper's dynamics use bounded memory when $\delta_m < 1$. In that constant-gain regime, the decreasing-gain convergence analogues no longer prove almost-sure convergence. The relevant formal target is instead a stationary distribution concentrated near the attracting regions, with escape times and residual dispersion set by opportunity and memory: high-N contrasts sit tightly near the extremes, while low-N or outside-option-dominated contrasts remain diffuse. That stationary distribution claim is highly likely ($p \approx 0.85$)[4] under an explicit condition that the outside option's fixed utility doesn't dominate the candidate set; without that condition, the direct transfer from the decreasing-gain analogues is much weaker.

Two consequences follow. First, selection intensity is proportional to $N_t(n, c) \cdot \Delta(d)$. Dense, update-configuring contrasts concentrate fastest. The categorical regime is expected for closed contrasts with many opportunities and high update divergence.

Second, low-Δ contrasts face weak selection, so their licensing distributions are predicted to stay more gradient: policed like style, not like grammar. The falsifier from §2 is derived rather than stipulated. Bounded memory adds a quantitative residual: even near an attractor, dispersion should scale with opportunity and forgetting, roughly with the effective concentration $N_t/(1-\delta_m)$.

Sampling drift near the extremes makes the boundaries sticky in finite populations, while middle states stay vulnerable to renewed evidence. The same escape-time logic connects winnerless-cell metastability (§4.6) and actuation (§4.8): a state can persist for long periods and then move quickly once

[4] Here and below, $p \approx \cdots$ marks calibrated confidence: how likely the informal claim is to survive formalization or direct empirical test, given the cited analogues and assumptions.

the evidence balance or outside-option utility changes. This connects the dynamics to complex-adaptive and cultural-evolutionary accounts of language change (Beckner et al., 2009; Blythe & Croft, 2012; Kirby et al., 2014), and to invisible-hand explanations, where macro-level order emerges as the unintended consequence of micro-level choices (Keller, 1994).

4.5 Derived obligatoriness

The state theory defines saturation relative to the obligatory dimensions $\mathrm{OBL}(c, A)$ (§3.2). Those dimensions aren't stipulated per language; obligatoriness is the limit of preemption applied to zero-marking.

> A dimension d belongs to $\mathrm{OBL}(c, \phi)$ for frame type ϕ iff, for every niche n in c with d at issue, every assembly leaving d unset has licensing driven to $C_t \approx 0$.

The derivation is the bimodality result applied with zero-marking as the disfavoured competitor. In a d-at-issue niche, the marked and unmarked assemblies compete; where the marked assembly wins every opportunity, the zero-marking assembly accumulates preemption mass exactly as any other rival does, and its posterior concentrates near zero. A dimension becomes obligatory once zero-marking has been lost in every niche where the dimension is at issue. The claim is a corollary, and it inherits the bimodality result's status–conditional on that transfer, roughly $p \approx 0.85$.

The typological payoff is that §2.1's descriptions convert to predictions. English progressive marking is obligatory because simple-present assemblies in ongoing-at-issue niches have been driven to near-zero licensing; French zero-marking there hasn't been, so the dimension stays optional, and *Elle étudie maintenant* is fine. Turkish evidential marking is the same story on a different dimension. And grammaticalization's obligatorification gets a mechanism: a contrast on its way to obligatory status is one whose zero-marking competitor is losing dense niches one at a time, which predicts an intermediate stage–obligatory in some frame types and niches, optional in others–that grammaticalization corpora are positioned to test.

4.6 Winnerless cells

Preempted gaps have a winning competitor. Defective paradigm cells don't, and the dynamics explain why the difference can be stable rather than transitional.

Suppose candidates $f_1, \ldots, f_m$ for a dense cell are near-symmetric in initial licensing, the outside option (periphrasis, avoidance) has moderate fixed utility, and producing a weakly-licensed form carries a face cost. Then the production rule routes the cell's opportunities to the outside option: no candidate accumulates positive evidence. Here the attributability weighting (47) matters: outside-option choices have $q_i \approx 0$, so no candidate accumulates much negative evidence either. Every candidate sits at low mean and low concentration–a self-maintaining interior state, metastable for long times, since escaping it requires some candidate to accumulate positive evidence that the avoidance attractor keeps starving it of.

The claim has two strengths. The winnerless-cell example in §3.9 relies on a qualitative mechanism: an avoidance attractor plus evidence starvation. The stronger metastability claim, persistence for times exponential in population size, is a conjecture with analogues in evolutionary dynamics, at perhaps $p \approx 0.6$ as stated and $p \approx 0.8$ for the qualitative mechanism. Sims's (2023) point that the maintenance mixture varies across cases–structural uncertainty, lexical restriction, learning dynamics, avoidance–is compatible with this narrower claim: the model supplies the avoidance-and-starvation component and is agnostic about the rest of the mixture.

The phenomenological contrast follows. Clean defective cells (*pobedit'* 1SG, with **amn't* depending on niche granularity) show low mean with low concentration–"I don't know how to say it"–where preempted gaps show low mean with high concentration–"that's impossible". Same mean, different concentration, different feeling, read out by Φ (§3.7). The satiation-framing prediction follows: defective cells should behave like low-evidence forms, movable under framing; preempted gaps shouldn't (Broadbent, 2009; Pertsova, 2016; Thoms et al., 2025).

4.7 Destabilization of moribund contrasts

Bounded memory yields one more result. With discount $\delta_m < 1$, the equilibrium concentration of a node's posterior is set by the balance of evidence inflow against forgetting: $\nu^* = O\big(N/(1-\delta_m)\big)$, with N the per-window opportunity count for the node's niches. If opportunity falls–a case distinction going moribund, a niche drying up–equilibrium concentration falls proportionally: licensing estimates drift toward their priors, and inter-speaker dispersion rises, because individual histories diverge when shared evidence thins. The discounted update makes this a high-confidence consequence ($p \approx 0.9$).

The corollary is a novel, testable leading indicator: for a contrast going moribund, judgment *dispersion* across speakers should rise measurably before mean judgments or production frequencies move. Dispersion leads; means lag. Variation corpora and judgment archives with repeated sampling over apparent time are well suited to check it, and §5.2 states it as a registered prediction.

4.8 Actuation

The same dynamics define ACTUATION: movement across the separatrix $C^\ddagger$ in (50). Actuation occurs when the expected positive stream $s_t(x,c)$ begins to outweigh the combined negative streams $e_t(x,c) + p_t(x,c)$ across successive windows, or when social and processing conditions lower β or raise α enough that the current state falls on the licensing-increasing side of $C^\ddagger$. Once that happens, the pivotal-node mean $C_{t+1}(\kappa_x, c)$ in (46), and hence $G_{t+1}(x,c)$, rises rather than decays, yielding the familiar S-curve at the population level.

The factors that drive such bifurcations align with the motivations discussed in §2.5. Semantic reanalysis may increase utility (and thus $\rho_t^\star$) by making a form–value relation more transparent or useful. Social pressures may shift utilities (prestige effects) or alter the relevant community standard. Structural motivations such as analogical extension can systematically raise licensing probability by borrowing strength from frequent neighbors. Processing innovations may reduce error rates (e_t) or locality costs, improving the net evidence balance.

Individual innovation is insufficient. A few speakers adopting a marginal construction don't guarantee its success; the evidence balance has to shift so that licensing systematically outweighs non-licensing across the relevant community. Many sensible innovations fail because they appear before those community conditions are in place.

Near the separatrix, small perturbations in community attitudes can trigger rapid shifts between exclusion and licensing. As the state moves further past $C^\ddagger$, reversal becomes increasingly unlikely. This produces the characteristic S-curve of documented language changes (Kroch, 1989). For constructions in the marginal zone, with the current state near $C^\ddagger$, the model predicts heightened sensitivity to external factors and greater cross-community variation. Small or peripheral communities may show volatile behaviour near bifurcation points before a change spreads to larger population centres.

The destabilization result sharpens the actuation picture at the other end of a form's life. Actuation is crossing a separatrix; destabilization is a collapse in concentration. A contrast can lose its grip when evidence of any kind stops arriving, without accumulating negative evidence; the model predicts the two routes are empirically distinguishable: actuation moves means along an S-curve with dispersion transiently high near the critical point, while moribundity raises dispersion first and moves means only as cohorts turn over.

4.9 Apparent exclusions from concentrated non-licensing

When a defined, saturated assembly exists but its pivotal node's licensing posterior is driven toward zero and concentrates there under persistent preemption in a dense opportunity set, speakers converge on a sharp, non-satiating rejection profile. Additional processing penalties in $R_i(u, c)$ (e.g. systematic garden-path repair due to an entrenched fused-head construal) can strengthen the subjective categoricality without any independent structural constraint being posited.

5 Theoretical implications

OVMG is not just a new decomposition of grammaticality. It says what the category should let us expect. If a form–value relation is grammatical in a communicative situation, it should pattern with other licensed resources in repair behaviour, transmission, satiation under exposure, and trajectories of change. If it's unlicensed, weakly licensed, or backed by little evidence, those expectations should fail in different, diagnostic ways (§3.3.2).

In this sense, the account is projectibility-first: grammaticality earns theoretical standing by the inferences it supports, not by the mere fact that a mechanism can be named for it (Boyd, 1999; N. Goodman, 1955; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). The profile that supports those inferences is conditioned form–value stability. The evidence offered here reaches three graded levels. First, bimodal licensing dynamics explain why some form–value relations stabilize near licensing or exclusion. Second, the production–repair loop explains how later use can help maintain that state. Third, the repair stream supplies a candidate controller: a process that registers departures from the licensed profile and pushes the community back toward it. That stronger claim isn't assumed; it depends on the intervention signature in §4.3 (Reynolds, 2026b; Woodward, 2001, 2003).

This recasts the competence–performance relation. Licensing and feeling are coupled layers, not a competence core plus performance noise: processing and other performance factors help shape which form–value relations stabilize, and those stable relations in turn constrain performance. But they are still separately identified. The identification strategy in §3.6 keeps this from becoming a catch-all, since licensing, anomaly, and confidence are measured by different evidence streams and can falsify each other. The rest of this section develops the consequences: the account's reading of macro-typological universals, its distinctive predictions, and its relation to neighbouring frameworks.

5.1 Macro-typological constraints on grammatical design

Recent macro-typological work sharpens the question of how strongly grammar is constrained across languages. Verkerk et al. (2025) test 191 implicational universals from the Universals Archive against Grambank's 2,430-language morphosyntactic sample, using Bayesian models that control explicitly for genealogical and areal non-independence and then follow up with spatiophylogenetic analyses of evolutionary rates.

A naïve analysis that ignores relatedness appears to support the vast majority of proposed universals, but once phylogeny and geography are accounted for, only about a third (60 of 191) remain statistically supported. Support is concentrated in relatively narrow domains: most hierarchical universals and a substantial minority of "narrow" word-order universals survive, whereas "broad" word-order universals and miscellaneous others fare poorly. Diachronically, supported universals typically correspond to "harmonic" combinations of features (for instance, consistent head–dependent orders) that languages are more likely to evolve into than out of, with these preferred configurations recurring independently across lineages.

From an OVMG perspective, these findings identify candidate attractors in the design space of operator-valued form–value relations rather than exceptionless grammatical laws. They don't by themselves show that a macro-typological feature is a language-internal grammatical object. The typological claim has to pass through a comparandum-to-realization mapping: one must specify which comparative feature is being tracked, how each language realizes the relevant operator value or form–value relation, and whether the resulting profile predicts held-out languages or diachronic trajectories (Reynolds, 2025).

Under that discipline, supported patterns correspond to configurations for which the net evidence balance (positive evidence s_t vs. preemption/error mass $e_t + p_t$) in the dynamics of §4.3 is positive across a wide range of communities. They're cognitively and communicatively favourable enough that repeated episodes of change tend to push the induced licensing posterior $C_t(\kappa, c)$ toward fixation in lineage after lineage.

The fact that roughly two-thirds of the tested universals fail once autocorrelation is controlled for also fits a de-idealized view of grammaticality. Community-specific conventions still have considerable freedom in how they realize operator values, with only some regions of the space strongly preferred. Large-scale comparative work of this kind complements OVMG by locating candidate stable regions; OVMG explains how local community dynamics and operator compatibility can make those regions reachable and persistent.

5.2 Predictions

The model's distinctive predictions come from separating positive evidence, opportunity structure, counterfactual choice probability, posterior concentration, and repair flow. Cross-linguistic gender remains a useful boundary case, but it isn't the strongest test. Any framework that treats grammaticalized concord as obligatory predicts stronger judgments where concord is more tightly grammaticalized. The companion operator-stratum account also complicates the simple gender story: gender concord can be policed through paradigm inertia even when its public-update value is attenuated (Reynolds, 2026c). The sharper predictions concern concentration contrasts, zero-evidence learning, L2 transfer, moribundity, repair policing, and change by re-licensing.

Satiation framing, as a rank order. Framing manipulations are prior shifts, and a prior shift moves a posterior in proportion to $1/\nu$–the inverse of its concentration. The prediction is ordinal across construction types, once $\rho_t^\star$ has identified the relevant niche: confirmed preempted gaps ≈ **sheeps* (nil) < entrenched-but-uncommon constructions < independent relative *whose* ≈ clean winnerless cells such as *pobedit'* 1sg (largest). Participants told that tokens of a low-concentration form come from an intentional dialectal, literary, or ingroup resource should show more improvement than participants told the same tokens are errors; the same manipulation should leave a confirmed preempted gap unmoved. Lu et al.'s (2024) meta-analysis of syntactic satiation in extraction from islands supplies a nearby benchmark for extraction phenomena, not a direct measurement of left-branch extraction. The dependent variables are rating change, confidence change (Φ), later repair behaviour, and elicited production.

Artificial-language learning. An artificial-language-learning experiment can hold positive evidence constant at zero while manipulating opportunity-set size. Learners could see a system in which a target candidate is never produced. In the high-opportunity condition, competitor candidates repeatedly occupy the relevant niche and the target has high $\rho_t^{\star}$; in the low-opportunity condition, the niche rarely arises or the target would have low counterfactual utility. OVMG predicts categorical rejection in the first condition and uncertainty in the second, even though both conditions contain identical zero positive evidence. The design operationalizes the Figure 4 contrast and is compatible with artificial-language learning paradigms that manipulate simplicity and communicative structure (Culbertson & Kirby, 2016).

L2 preemption transfer. Early L2 rejections should track L1 preemption mass, not just L2 frequency. A learner whose L1 strongly preempts a candidate in a high-opportunity niche should reject the L2 analogue early, even if the L2 input frequency is low enough to leave native speakers uncertain. Conversely, low L1 preemption should make learners more willing to treat rare L2 patterns as unresolved rather than impossible. As learners join the L2 speech community, $q_h(c \mid e)$ and the constructional hierarchy should shift toward the target norm-centre. This view aligns with Selinker's (1972) concept of interlanguage, but gives a more specific prediction: transfer is strongest where the L1 has accumulated large effective preemption mass.

Dispersion as a leading indicator. For a contrast going moribund, the destabilization result (§4.7) predicts that inter-speaker judgment dispersion rises measurably *before* mean judgments or production frequencies move: falling opportunity thins the shared evidence, concentrations decay, and individual histories diverge while the average still holds. Dispersion leads; means lag. Variation corpora and judgment archives with repeated sampling over apparent time can test this directly, and the prediction is distinctive: accounts on which decline is driven by accumulating negative evidence predict means and dispersion moving together.

Policing intensity as a parameter. The repair stream (§4.3) converts the operator/style boundary from a classification into a regression: policing intensity should scale with the product of niche opportunity and update divergence, $N_t \cdot \Delta$, with both factors independently estimable–opportunity from opportunity-annotated corpora, divergence from comprehension probes or annotated update-divergence tasks fixed before the repair data is inspected. The §2 falsifier becomes quantitative: a closed, dense, non-configuring contrast has $\Delta \approx 0$ and should show style-like policing–a social-sanction baseline with gradient, unimodal licensing–no matter how frequent it is. If repair rates track social indexing rather than Δ, the corrective term collapses into sociolinguistic policing and the controller claim fails.

Change by re-licensing. Hard unification makes a commitment that a weighted-penalty model doesn't: no amount of entrenchment can make the same clashing operator assignment grammatical. Apparent change in cases like **I've finished it yesterday* should come either from read-out effects or from re-licensing. Where present-perfect combinations with definite past-time adverbials are attested in a community (Werner, 2013a, 2013b), change has to proceed by *re-licensing*: a perfect construction with a different operator contribution enters the repertoire, predicting judgment distributions that are bimodal across communities and categorical within speakers, not gradual within-speaker reduction in the status cost of the original clash. A soft-constraint model predicts the opposite signature. Judgment-distribution data from the relevant contact varieties decides it.

This contrasts with preempted gaps. A preempted gap can re-open when the evidence balance changes sign: the pivotal node's mean rises through the middle, producing within-speaker gradience during the transition and a community S-curve. Compatibility failures can't re-open that way; they require a distinct construction with a different ω. The identification discipline is the same as elsewhere: the re-licensed construction has to be diagnosed before the judgment data is used, for example by showing that it supports other definite past-time adverbials, scope interactions, or ellipsis patterns predicted by the new operator contribution.

Intonational operators. Nothing in the model privileges segmental material, so intonational patterns are predicted to count as operator exponents wherever they form closed, high-opportunity, update-critical repertoires: the English polar-question rise and nuclear-accent placement configure what an utterance does to the common ground much as tense and agreement do (Reynolds, 2026c). The Turkish suffix-harmony analysis in Appendix A gives the parallel case for segmental exponent choice: phonological material matters for grammaticality when it realizes an operator exponent. The corresponding intonational test is whether mismatches between intonation and text (e.g. nuclear accent on a closed-class word under broad focus) are policed like grammar, with categorical rejection, repair, and resistance to framing, or like style.

5.3 Relationship to generative grammar

Generative work made a central contribution by showing that grammaticality can't be reduced to semantic plausibility, processing ease, or frequency of attestation. Chomsky's *Colorless green ideas sleep furiously* made the contrast visible: speakers can recognize syntactically well-formed sentences even when they're semantically bizarre. Generative analyses explain categorical constraints like the impossibility of extracting determiner-adjective sequences in English (**Which do you prefer car?*), and they capture the fact that such sequences remain ungrammatical even when their intended meaning is clear and processing demands are low.

OVMG retains the lesson. Judgments reflect systematic patterns (Dennett, 1991); those patterns can't be reduced to meaning or processing alone; and certain syntactic configurations appear to be excluded regardless of context. The earlier diagnostics for center embeddings and preempted gaps build on ideas about recursive structure and acknowledge the generative discovery that some exclusions are systematic. They relocate the explanation in coverage, hard compatibility, saturation, and concentrated exclusion.

The same point governs the account's use of formal evidence. A generative analysis can supply a real warrant when it specifies the target, diagnostics, evidence, and failure conditions for a grammatical contrast. OVMG isn't an attempt to replace every I-language analysis; it rejects only the stronger claim that categorical grammaticality must be explained by an autonomous syntactic component or a UG-backed inventory of primitives. Its target is the community-level licensing profile that makes some form–value relations categorically policed.

One well-known case is independent relative *whose* (1d). As Hankamer and Postal (1973) observe, the construction appears to violate no syntactic principles: independent genitives are possible (*Mine was visiting*), independent interrogative *whose* is grammatical (*Whose was open?*), and *whose* can be used as a dependent relative pronoun (*the student whose friend was visiting*). The generative tradition's careful documentation of such cases, where seemingly parallel constructions show puzzlingly different grammatical status, has driven theoretical development.

OVMG explains the contrast differently. Rather than positing an autonomous syntactic component, it treats grammatical constraints as products of form–value relations within specific communicative situations. For independent relative *whose* to be felicitous, multiple conditions have to converge: the possessor has to be sufficiently accessible in the discourse while the possessum is predictable enough to license ellipsis, but the possessive relationship needs to be semantically significant enough to warrant explicit marking, and this configuration has to occur in a context where a relative clause is the natural way to package this information.

The relevant niche is tiny. Speakers encounter the construction so rarely, despite perfectly common components, that the licensing posterior remains diffuse; even when all conditions align, the construction feels alien because the evidence is thin, not because the community has accumulated confident negative evidence against it.

The difference is evidential rather than merely terminological. A generative theory has to explain why a syntactically possible and pragmatically useful construction remains marginal. OVMG predicts the profile of that marginality: low concentration, framing sensitivity, and weak confidence, not the non-satiating rejection profile of a preempted gap. The result preserves the generative recognition of systematic constraints while embedding it in a theory of how form–value relations become established and maintained in language communities.

Other cases that generative grammar struggles to explain are grammatical with one meaning but ungrammatical with another, such as *have*+numeral years. While syntactically identical to grammatical expressions like *I have 25 dollars*, this construction becomes ungrammatical specifically when used to express age. A purely syntactic account would have to explain why the same structure is well-formed in one case but ill-formed in another, despite no apparent syntactic differences.

OVMG, in contrast, locates the source of ungrammaticality in the community's form–value conventions: *have*+numeral years is licensed for duration or future time (*I have 16 years until retirement*), but the age-predication use is driven toward zero in English by dense competition with the *be*+age frame (*I am 16 years old*). Similar cases arise with plural forms that are grammatical with some meanings but not others (e.g. *peoples* for ethnic groups but not multiple individuals) and with verbs that resist certain arguments despite no obvious syntactic prohibition (e.g. *discuss about*). These meaning-dependent grammaticality patterns suggest that licensing is keyed to specific form–value associations rather than purely structural templates.

5.4 Relationship to Construction Grammar

Construction Grammar (CxG) (Fillmore, 1988; Fillmore et al., 1988; Goldberg, 1995, 2019; Kay & Fillmore, 1999; Sag, 2012) made a central contribution by treating learned form–meaning pairings as grammatical objects. At its core, CxG argues that language consists of learned relationships between form and meaning at multiple levels of complexity. These form–value relations, or constructions, include both individual morphemes and abstract syntactic patterns. This perspective helps explain phenomena that proved challenging for earlier approaches, including idiomatic and partially schematic expressions that don't fit a clean core/periphery split.

CxG shows that meaning suffuses all levels of grammatical organization. Rather than treating syntax as an autonomous formal system that interfaces with semantics only at designated points, CxG reveals how meaning and form are inseparable aspects of linguistic knowledge. For instance, the *What's X doing Y?* construction carries an implication of incongruity that can't be derived from its component parts (Kay & Fillmore, 1999). Recent work connects this framework to model-internal analysis: Weissweiler et al. (2023) use construction grammar to probe how neural language models handle different levels of linguistic abstraction.

OVMG shares these CxG commitments about form–value relations and constructional abstraction. It doesn't privilege one expressive substance over another. The companion operator-stratum paper explicitly denies a substance-based boundary: phonological, gestural, lexical, morphological, and syntactic material can all realize operator contrasts when they enter a closed, accountable public-update repertoire (Reynolds, 2026c). The OVMG claim is narrower. Categorical grammaticality clusters around form–value relations with a particular role: high-opportunity, closed-paradigm contrasts that configure update, role allocation, scope, reference, or repair.

The contrast becomes clear when speakers judge violations. Register, politeness, genre, and many lexical choices can be stable and socially important without blocking the public update itself. A wrong tense, agreement value, or clause-linking marker can mis-set the update instruction; a wrong obligatory allomorph can instead fail as an unlicensed exponent choice for a closed operator. The age-predication example (*I have 25 years*) is transparent but not licensed for that English frame; its rejection comes from non-licensing in a high-opportunity constructional niche, not from the substance of syntax or morphology as such.

OVMG preserves CxG's systematic treatment of constructions while adding a prediction about which constructional contrasts attract categorical policing. The relevant contrasts are those whose closure and opportunity structure drive the licensing posterior toward the bimodal regimes in §4.4. That lets CxG's continuum of constructions coexist with a non-arbitrary explanation of why some violations are heard as grammar and others as style, stance, or appropriateness.

5.5 Relationship to usage-based approaches

OVMG shares with usage-based approaches (UBA) (Bybee, 2006, 2007, 2010) the idea that linguistic knowledge develops from patterns of actual language use rather than from an autonomous formal system. Both perspectives reject the notion that grammaticality can be reduced to abstract rules operating independently of meaning and context.

Several pieces of the formal core come directly from this tradition. The speaker/population split restates Schmid's (2020) separation of ENTRENCHMENT, the cognitive reorganization of a speaker's knowledge, from CONVENTIONALIZATION, the social establishment of a community's regularities. Here $z_{j,t}(\kappa, c)$ gives speaker-level inclusion states, and $\theta_t(\kappa, c)$ with posterior mean $C_t(\kappa, c)$ gives the population licensing rate. The population dynamics themselves descend from Croft's (2000) utterance-selection theory, in which variants are replicated and selected within a community; the S-curve results the model reuses are already worked out in that lineage (Blythe & Croft, 2012).

Frequency per opportunity is likewise not new: it's the core of Stefanowitsch's (2006) case that corpora contain negative evidence, where a form counts as "significantly absent" rather than "accidentally absent" only once its observed frequency is weighed against the frequency expected given its opportunities. Variationist sociolinguistics has normalized by opportunity since Labov's principle of accountability (Labov, 1972).

OVMG adds their joint formalization and a prediction. The licensing-versus-selection factorization $\pi_t \approx G_t \cdot \rho^\star_t$–or $C_t(\kappa, c) \cdot \rho^\star_t$ in single-pivotal-node cases–turns Croft's replication-versus-selection into an estimable product. The gated preemption mass $p_t = \sum_i r_i q_i \delta_m^{t-t_i}$ reduces to $N_t \cdot \rho_t^\star$ in the high-confidence competitor regime, putting a quantity on the absence that Stefanowitsch left open, since he showed how to establish that a form is significantly absent but stressed that significant absence "does not, in itself, provide any clues as to why". Licensing-versus-selection is exactly that missing clue: a significantly absent form may be unlicensed ($G_t \approx 0$, or pivotal $C_t \approx 0$) or licensed but dispreferred ($\rho_t^\star \approx 0$), and the two project differently.

The Beta dynamics then convert these quantities into the bimodality result (§4.4) and the prediction about which contrasts attract categorical policing. The relation to Goldberg's (2019) COVERAGE and statistical preemption is one of formalization in the same spirit: her "explain me this" puzzle (a high-coverage double-object frame preempted by the prepositional dative) is a preemption-mass story in which the competitor wins a high-opportunity niche.

One caution follows from Ambridge et al. (2015), who find that overall form frequency (entrenchment) predicts the retreat from overgeneralization more robustly than competitor frequency (preemption), and who propose collapsing the two into a single competition process. OVMG keeps them analytically separate: error evidence e_t and preemption mass p_t are distinct streams, but they enter the Beta update additively (§4.3). The model can represent Ambridge and colleagues' single-process view as their sum while leaving the separability an empirical question rather than a stipulation.

The analysis of independent relative *whose*, as in ?*I saw Joan, a friend of whose was visiting*, is a case in point. A simple UBA account might predict that this construction's marginality stems from its low frequency. But this explanation proves insufficient: the construction is rare, and it's dramatically rarer than one would expect given the frequency of its component parts. Independent *whose* appears in interrogatives (*Whose is that?*), and the relative *whose* is common in dependent contexts (*the student whose paper was late*). Given these frequencies, analogical extension should make the independent relative more common than observed.

Raw component frequency is the wrong predictor; *whose* isn't a preempted gap. The frequency of independent and relative *whose* taken separately doesn't create opportunities for the niche that requires both at once, together with an accessible possessor, a predictable possessum, and a licensing ellipsis site; that convergence is vanishingly rare. This isn't the significant absence of a preempted form, which needs a large opportunity set and a competitor that keeps winning; it's the low-opportunity uncertainty described in the diagnostic categories above. The tension between "rarer than expected" and "uncertain rather than rejected" dissolves once licensing is separated from selection: *whose* is under-*observed* because its niche rarely arises, not preempted because a rival repeatedly beats it.

OVMG doesn't exempt the operator stratum from usage dynamics; it makes those dynamics more specific. A rare form can remain grammatical when the opportunity set is tiny or when higher-grain constructions license the token by compositional inheritance. A transparent candidate can become categorically rejected when the opportunity set is large, the analogical candidate would have had high $\rho_t^\star$, and an entrenched competitor wins every time. The result preserves usage-based findings about frequency and entrenchment while predicting when frequency becomes preemption.

5.6 Relationship to logicality of language accounts

An alternative perspective, prominent in recent formal semantics, suggests certain types of unacceptability stem from the language faculty itself possessing a deductive system that identifies and filters sentences with logically trivial meanings (tautologies or contradictions) (cf. Chierchia, 2013; Del

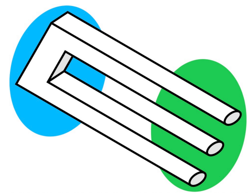

Figure 5: Impossible trident

Pinal, 2019; Fox, 2000). This "logicality of language" hypothesis attempts to explain, for example, systematic restrictions on quantifiers by arguing the unacceptable cases are "L-trivial", meaning their triviality arises solely from the meaning and configuration of logical or closed-class terms (like *every*, *some*, *not*), irrespective of the open-class words (like *student*, *run*) (Gajewski, 2009).

A key challenge is explaining why simple tautologies or contradictions (e.g. *It's raining and it isn't raining*) are often acceptable. Del Pinal (2019) argues against "Logical Skeletons" (which assume the system ignores open-class word identity) in favour of "LF+RESCALE", a view where the system sees standard logical forms but allows optional, context-dependent modulation of open-class terms (e.g. interpreting the second "raining" as "raining hard") to yield non-trivial meanings.

OVMG offers a broader account of ungrammaticality. "Logicality" approaches explain restrictions tied to logical and closed-class vocabulary through L-triviality. OVMG also covers cases less easily reducible to logical contradiction, such as absent form–value relations (1a), strong deviations from conventional community forms (17), and extreme, unexpected rarity (1d). By grounding grammaticality in community-specific conventions while separating hard compatibility from posterior licensing, OVMG accommodates cross-linguistic variation and rating gradience, which are less central to the L-triviality filter.

LF+RESCALE provides a specific mechanism for acceptable "trivialities". OVMG treats those cases as possible products of more general interpretive principles within community norms, without positing a dedicated deductive module or specific operators like RESCALE. It also keeps conventional grammatical status separate from subjective processing effects or "feelings" (§3.7). OVMG frames grammaticality as an emergent consequence of communicative practice in the formal sense of §4.4, rather than a direct output of logical computation within syntax.

5.7 Relationship to relevance-theoretic accounts

Recent work by Scott-Phillips (2026) makes a useful observation: linguistic intuitions about acceptability arise as byproduct effects of cognitive systems for interpreting communicative acts. Just as people immediately sense when a visual stimulus violates core assumptions about physical objects (as with impossible objects), they detect when utterances violate basic presumptions about communicative efficiency. Scott-Phillips argues that unacceptability occurs not from mere inefficiency, but from an inherent impossibility of interpreting an utterance consistently with these presumptions, much as an impossible trident (Figure 5) can't be interpreted as physically cohesive in any context.

OVMG shares several premises with this account. Both reject the need for an innate grammar faculty, locating linguistic intuitions instead within general cognitive systems, and both treat language as developing from communicative needs rather than autonomous syntactic principles. OVMG's emphasis on situation-specific form–value relations builds directly on Scott-Phillips's arguments about how communicative pressures shape linguistic conventions.

The frameworks differ primarily in their explanatory mechanisms. Where Scott-Phillips argues that grammaticality judgments reduce to impossibilities of efficient interpretation, OVMG suggests that while communicative pressures shape which form–value relations become conventionalized, these relationships then create systematic constraints that can't be reduced to efficiency alone. For Scott-Phillips, one would have to demonstrate that (1c) contains inherent contradictions making efficient interpretation impossible. OVMG instead analyzes how *yesterday*'s deictic temporal anchoring fails to unify with the present perfect's reference-interval contribution. While these analyses might ultimately converge, it remains unclear why the criterion of inherent impossibility of interpretation should apply specifically to operator-value violations rather than to lexical-lexical conflicts or certain phonological patterns. The scope of what constitutes an interpretive impossibility requires further theoretical development.

Empirically, OVMG requires tests of whether specific form–value relations are stable within a communicative situation and where operator-value conflicts arise. The challenge for relevance-theoretic accounts, as Scott-Phillips (personal communication, Dec. 16, 2024) acknowledges, lies in establishing independent, empirically vulnerable claims about what makes efficient interpretation inherently impossible rather than merely difficult. The clearest comparison would test the two accounts on novel constructions as they become acceptable or unacceptable.

6 Limitations and future directions

The account has real limits, and each points to work it invites rather than forecloses. One is its reliance on the concept of "communicative situation". Speakers move among overlapping contexts that shift with the immediate setting, durable social positions, and the norms they orient to. The account builds in this fluidity, but operationalizing these dimensions, and measuring their influence on grammatical stability, will take more precise methods. A further limit is mathematical candour: the bimodality claim rests on close decreasing-gain analogues and on a bounded-memory stationary-distribution conjecture with an explicit outside-option condition (§4.4); the framework's central emergence claim inherits that provisional status until the bounded-memory case is proved.

A second boundary concerns stylistic variation and individual preference. Most choices that fall within grammatical acceptability aren't cases of (un)grammaticality at all. They matter for this account only when they become tied to communicative situations strongly enough to affect licensing, repair, or categorical rejection. The account still has to explain how stylistic preferences become grammatical licensing facts.

A third concerns the boundary of constructional coverage. The paper argues that many apparently arbitrary restrictions are either compatibility failures, saturation failures, or concentrated non-licensing under historical preemption. Cases that resist those treatments should be analyzed as failures to build a well-typed covering assembly, or else as evidence that the constructional inventory has been specified too narrowly.

6.1 Methodological considerations

The development of experimental syntax since the 2000s (Schütze, 2016) has transformed how gradient acceptability is measured, letting researchers quantify subtle differences in judgments and track how they change with exposure. The framework's predictions invite corpus-based, experimental, and cross-linguistic investigation. Several avenues stand out:

1. *Corpus analysis:* Large, balanced corpora allow researchers to track frequency patterns and stability over time. Investigating rare or marginal forms (e.g. the independent relative *whose*) in corpora can reveal whether low frequency corresponds to genuinely unstable operator-value patterns or merely a sampling artifact. Comparative corpus research in multiple languages can test predictions about how community values influence grammatical distinctions.
2. *Judgments:* Controlled psycholinguistic tasks, including magnitude estimation or forced-choice paradigms, can assess rating gradience, confidence, and test–retest stability. Exposure and familiarity manipulations measure satiation effects and distinguish entrenched constraints from constructions that become more acceptable through repeated exposure. Exposure effects should be interpreted through the two-channel read-out: rating improvement without corresponding change in confidence, production, or repair behaviour is evidence for movement in the anomaly channel rather than for a change in licensing status.
3. *Processing measures:* Eye-tracking, self-paced reading, and EEG studies can identify whether some judgments arise from processing overload rather than durable grammatical constraints. If a form becomes easier to parse with practice while production and repair indicators remain stable, the effect belongs to the implementation-cost channel rather than to the licensing posterior.
4. *Sociolinguistic fieldwork:* Investigations in communities with distinct dialects or multilingual practices can clarify how social and pragmatic motivations shape grammatical stability. Elicitation tasks and careful comparisons across speech communities can confirm that what counts as grammatical depends on local norms rather than universal principles.
5. *Longitudinal and historical studies:* Diachronic corpora and historical grammars can trace how marginal constructions evolve, testing predictions about which factors lead certain patterns to stabilize, which fade, and which undergo reanalysis. Tracking a community's licensing of previously dubious constructions over time provides direct evidence for the gradual dynamics posited by the framework.
6. *A decision-layer discontinuity check:* The threshold $\tau(c)$ is a read-out parameter, not a constitutive boundary. A discontinuity in response type near $\tau(c)$ would test the decision layer: whether asymmetric losses shift hearers from acceptance to repair, request for clarification, or sanction, over and above the smooth effect of G_t. The general design is a regression-discontinuity problem (Calonico et al., 2014; Lee & Lemieux, 2010); multinomial and covariate-adjusted versions are the needed extension (Lazzaro, 2026). In a confirmatory test, $\tau(c)$ has to be estimated and registered before the outcome data is touched, and the bimodality result (§4.4) predicts thin support near the threshold. Smoothness through $\tau(c)$ would weaken the proposed decision-layer switch, not the licensing-state theory itself.

Figure 6 states the decision-layer check in one picture: a discontinuity is evidence for a response switch at $\tau(c)$, while a smooth curve is evidence that the read-out tracks the posterior mean continuously.

7 Conclusion

Grammaticality, de-idealized, isn't a categorical fact about an ideal speaker in a homogeneous community but a conditioned form–value relation. Its profile–whether some assembly covers the form, whether the assembly's operator contributions unify, whether its obligatory dimensions are saturated, and whether the population licenses its constructions–is specified relative to a communicative

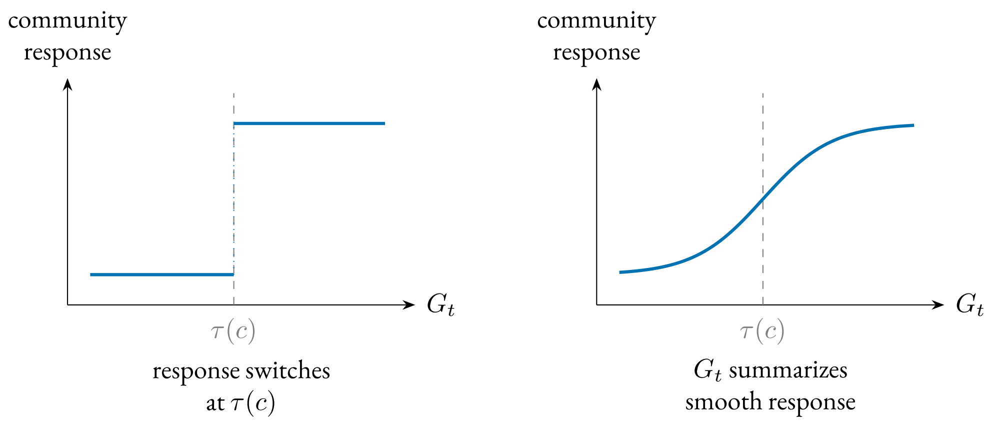

Figure 6: The decision-layer discontinuity check. If asymmetric losses induce a response switch, community treatment of a form should change discontinuously as G_t crosses the situational threshold $\tau(c)$ (left); if the read-out tracks the posterior mean continuously, treatment should vary smoothly through it (right). A regression-discontinuity design on multinomial hearer responses targets this decision layer, not the constitutive status state itself.

situation and read through a posterior that carries concentration as well as mean.

The competence–performance split becomes two layers with separate evidence streams, a conventional, population-level status and a two-channel subjective read-out–anomaly and confidence–identified by different measures so that they can falsify each other (§3.6). Where licensing is at issue, categoricality is then derived rather than stipulated: an emergent bimodality of the licensing state under preemption (§4.4). Coverage failure and hard compatibility remain distinct routes rather than default explanations for every sharp rejection.

What the profile is for is projection. A category earns its standing by supporting expectations licensed by partial observation, not by the mere presence of mechanisms: repair, transmission fidelity, satiation, and trajectories of change, over the licensing state as such, its concentration as well as its mean (§3.3.2).

The supporting claims are graded separately. Bimodal dynamics stabilize the profile, and the production–repair loop maintains it. The repair stream supplies a candidate corrective system–a process that registers operator mis-settings and pushes the licensing state back–and states the intervention signature that would earn it controller status, along with the finding that would defeat it (§4.3). The stronger claim is thereby made testable rather than merely left open. On this account, grammaticality matters because it predicts which forms will be repaired, transmitted, softened by exposure, or resistant to change.

These commitments expose the framework to failure. The sharp tests aren't exhausted by the decision-threshold design but form a family of profile-sensitive contrasts: confidence should dissociate preempted gaps from clean winnerless cells at matched mean ratings; framing effects should follow the inverse-concentration rank order; artificial-language learners should distinguish zero evidence from strong preemption; early second-language rejections should track first-language preemption mass rather than second-language frequency alone; moribund contrasts should show dispersion rising before means move; policing intensity should scale with $N_t \cdot \Delta$; and compatibility failures should change by re-licensing, not by gradual within-speaker cost reduction.

The account doesn't have the simplicity of a one-variable theory. It doesn't reduce grammaticality to a single scalar, a competence–performance split, or a social-pragmatic explanation. Its economy is architectural: the same recurring distinctions (coverage, compatibility, saturation, licensing, subjective read-out, opportunity structure, and diachronic support) handle cases that otherwise invite separate stories. That economy has to be earned empirically. The conditions need independent indicators, and the projections need to fail when those indicators are wrong.

It also carries a significant memory burden. Speakers need not store every utterance as a separate item, but the account asks them to retain many situation-indexed pairings among forms, operator values, constructional neighbours, opportunity estimates, competitors, and repair histories.

That inventory is large, but the framework makes it the model's data structure: the situation-indexed pairings among forms, values, neighbours, opportunity estimates, competitors, and repair histories are exactly the sufficient statistics the licensing posterior runs on (§§3–4). Compositional inheritance across assemblies keeps first-heard tokens from starting at zero, and future hierarchical pooling across construction families, lexemes, and situations would make the burden still more estimable. If linguistic memory is presumed small in advance, stored generalizations get forced into a rule-plus-exception format, and conditioned licensing is misdescribed as residue.

The program these predictions define is measurement-first. Estimate licensing from converging non-rating indicators (corpus rate per opportunity, elicited production, repair), fix the conditioning state before predicting rather than after, and prioritize cross-linguistic contrasts whose operator status differs, since that's where the account predicts categoricality should form or dissolve. De-idealizing grammaticality trades a tractable idealization for a profile that answers to evidence. It yields a category that predicts which rejections soften, which forms transmit, and which absences hold.

A Turkish vowel harmony and operator-exponent realization

Turkish illustrates a sharp distinction between LEXICAL disharmony inside stems and ALLOMORPHIC harmony on inflectional suffixes.[5] Only suffixal harmony matters for OVMG, and even there the role is indirect. Vowel backness isn't a public-update value; plural and past are. In Turkish, the exponents of those operators have to satisfy suffixal harmony.

STEM-INTERNAL VOWELS: DISHARMONY TOLERATED. Loanwords such as *doktor* 'doctor' violate backness harmony, but are fully acceptable; speakers store the form, and the community-level licensing state remains near 1.

(51) doktor
doctor
'doctor' (disharmonic stem, grammatical)

Because the word contains no malformed operator exponent, a stored construction covers it, the relevant operators unify, and $G_t \approx 1$.

SUFFIXAL HARMONY: OPERATOR-EXPONENT LICENSING. Inflectional morphemes are lexicalized with an underspecified vowel; the licensed allomorph copies stem backness and, for some suffixes, rounding. Using the "wrong" vowel doesn't make the intended value unrecoverable: speakers can identify *-ler* as a plural exponent and *-du* as a past exponent. What fails is licensing of that exponent choice in a dense operator-exponent niche.

(52) a. kitap-lar
book-PL.BACK
'books' (harmonic, grammatical)

b. * kitap-ler
book-PL.FRONT
intended 'books' (suffix harmony violation)

(53) a. gül-dü
laugh-PST.FRONT.ROUND
's/he laughed' (harmonic, grammatical)

b. * gül-du
laugh-PST.BACK.ROUND
intended 's/he laughed' (suffix harmony violation)

OVMG ACCOUNT. For the ill-formed **kitap-ler* and **gül-du*:

- The plural and past constructional nodes are licensed at the population level, $C_t(\kappa, c) \approx 1$, with speaker-level inclusion states correspondingly concentrated near inclusion. What is not licensed is the mismatched exponent-choice node.
- The mismatched allomorph is still identifiable as an intended exponent, so the token is covered by a candidate operator-exponent assembly. But its pivotal exponent-choice node is licensed to near zero by overwhelming preemption from harmonic suffix choices across the dense plural and

[5] See (Kabak & Vogel, 2001; Sezer, 1981) for discussion of the phonology and (Arık, 2015) for experimental evidence on native judgments.

past opportunity set. Thus $G_t \approx 0$ with very high ν: a licensing failure with high Φ, not empty coverage.

So speakers judge the word categorically wrong, not just odd-sounding, and the confidence is evidential rather than structural: the posterior has concentrated because every relevant suffix opportunity teaches the same allomorphic choice. This keeps suffix-harmony errors apart from word salad. **Can the have running* lacks a covering assembly; **kitap-ler* has a transparent intended value and an identifiable exponent, but the exponent-choice node is licensed to zero. By contrast, *doktor* in (51) is covered by its own stored node: stem disharmony violates no operator-exponent licensing condition, so $G_t \approx 1$ and the disharmony registers only as a phonological cost in R_i (§3.7).

Localized exceptions. Some derivational suffixes (e.g. *-imsi* '-ish') are lexically marked "disharmonic". Speakers store them, and community-level licensing remains near 1: no exponent condition is violated, so $G_t \approx 1$ despite vowel mismatch. Clitic boundaries that start a fresh harmony cycle (Kabak & Vogel, 2001) are handled the same way: they satisfy the operator-exponent requirement and leave harmony to phonology alone.

Turkish suffix-harmony errors are grammaticality failures in a precise sense: they don't mis-set a public-update contrast themselves, but they choose an unlicensed exponent for a closed inflectional operator in a dense niche. Stem disharmony lowers phonological well-formedness and affects only the subjective cost vector. A counterexample would be a case where phonology alone produced categorical, framing-resistant rejection without any operator-exponent licensing requirement in play.

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